

Supporting Information: matrix-free voxel elasticity

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This supporting document accompanies the matrix-free voxel elasticity study. It contains operator construction, methods, accuracy, outcome, capacity, comparator, audit and hypothesis evidence. Numerical regeneration and selected CPU checks use retained records; no new GPU performance experiment is represented. Source paths are relative to the accompanying artifact. CSV/JSON files retain exact per-observation values and references where PDF tables use rounded summaries.

Algebra, quadrature and counter methods

This supplement specifies the operator construction and measurement definitions. Its numerical work consists of CPU algebra and reanalysis of retained counters; it contains no fresh GPU measurements.

Explicit parity permutation and coefficient convention

Let the native Hex8 node order be (000,100,110,010,001,101,111,011), and encode a binary node as $n = n_x + 2 n_y + 4 n_z$. Let $p=0, \dots, 7$ label coordinate modes and $c=0, 1, 2$ label the x, y and z displacement components. Define

$$H_{pn} = (-1)^{\text{popcount}(p \& n)}, \quad T_{3p+c, 3i+d} = H_{p, n_i} \delta_{cd}, \quad TT^\top = 8I.$$

Thus T includes the native-to-binary node-column permutation and its output uses (p, c) order, with flat index $3p+c$. The block order is (g, c) , with $g = p \text{ xor } 2^c$ and flat index $3g+c$. The permutation matrix is

$$P_{3g+c, 3(g\oplus 2^c)+c} = 1$$

with all other entries zero. Write $A = T K_0 T^\top$. Each block is completely specified by

$$(C_g)_{cd} = A_{3(g\oplus 2^c)+c, 3(g\oplus 2^d)+d}, \quad c, d \in \{0, 1, 2\}.$$

For the stated element class,

$$C = PTK_0T^\top P^\top = \text{diag}(C_0, \dots, C_7), \quad K_0u = \frac{1}{64}T^\top P^\top CPTu.$$

The factor $1/64$ arises because $T^\top(-1) = T^\top T/8$ appears on both sides of the transformed matrix. The implementation stores $C_g/64$, not C_g : `code/hex_modal/modal.py` first forms `full=T@K0@T.T/64`, then selects `GROUPS[g,c]=3*(g xor (1<<c))+c`. Consequently its two unnormalised butterflies need no final division. In the eight-lane CUDA implementation, a lane owns mode p for all three components; the x/y/z coefficient blocks have group indices $p \text{ xor } 1$, $p \text{ xor } 2$ and $p \text{ xor } 4$. The required partner for component d in output component c is mode $p \text{ xor } 2^c \text{ xor } 2^d$. This implements P and $P^\top$ through indexing/shuffles without materialising a permutation array.

The exactness class is axis-aligned rectangular trilinear Hex8 elements, with an isotropic elasticity tensor constant within each element. Density multiplies the element tensor by a scalar modulus; it does not alter this symmetry. Under reflection of coordinate a , the scalar coordinate mode has character $(-1)^{p_a}$, while component c receives the additional vector sign $(-1)^{[a=c]}$. The vector-mode character is therefore $(-1)^{p_a + [a=c]}$. The elastic energy bilinear form is invariant under each reflection. For two modes with different characters, applying that reflection changes the sign of their coupling while preserving its value, so their coupling must vanish. Equality of all three characters is precisely equality of $g = p \text{ xor } 2^c$. Each of the eight characters has three component-mode pairs. All 24 modes are retained. This argument does not extend the exactness claim to distorted or sheared cells, arbitrary rotations with untransformed material/components, anisotropy or material variation within the cell.

The fresh check uses $E=1$, $\nu=0.3$, exact $2\times 2\times 2$ Gauss integration and FP64. Reconstruction means $\|K_0 - K_{\text{reconstructed}}\|_F / \|K_0\|_F$; off-block means $\|C - \text{blockdiag}(C)\|_F / \|C\|_F$. Eigenvalues with absolute value at most $1e-10$ $\|K_0\|_2$ are counted as zero. The six analytic translations/rotations are separately checked as a null basis, and 24 coordinate vectors plus 32 seeded normal vectors check the implemented product.

Cell side lengths	Reconstruction relative Frobenius error	Off-block relative Frobenius norm	Near-zero / positive / negative eigenvalues	Smallest positive eigenvalue
1×1×1	2.3077393e-16	1.6064287e-16	6 / 18 / 0	0.06410256
1×2×3	1.7818459e-16	1.3207403e-16	6 / 18 / 0	0.02548896

The reconstruction errors are approximately $2.31e-16$ and $1.78e-16$, respectively. All blocks, eigenvalues, actual thresholds and additional residuals are generated in [algebra-checks.json](#); the compact manuscript table is [algebra-invariants.tex](#).

The local plain, sign-bit and signed-multiply-add butterflies agree bitwise on the tested CPU FP64 corpus under the tested round-to-nearest arithmetic assumptions, including signed-zero and subnormal inputs. Multiplication by ± 1 is exact in this scope, which explains the local identity. The CPU emulation is not a universal compiler/rounding/NaN guarantee. Complete assembled GPU products include floating-point atomic scatter, whose order may differ, and are described as numerically equivalent within the declared tolerances. They are not claimed bitwise identical.

The source comparison must identify its scope. In `experiments/session_e/dependencies/cuda.py`, `modal8_muladd` and `modal8_signbit` are generated from `modal8` by changing the transform body and names; the layout, block product, modulus scaling and atomic scatter remain the same source text. The existing source-equality test checks precisely this. These variants deliberately change floating-point instruction selection. The `dense8` control in the same file uses the same eight-lane mapping but a dense local matrix product, and therefore does not differ from parity only in index arithmetic. The frozen code `hex_modal_cuda/cuda.py` is the earlier plain/dense prototype.

Composite midpoint quadrature in the pinned release

The retained source observation identifies `getKESubspace` in the pinned OpenMP GPU release (b46ed453) as using ten midpoint samples per fine-subcell axis. At the fine level this is **composite 10×10×10 midpoint quadrature per fine element**, namely 1000 samples. The coarse-level construction integrates its fine subcells with the same per-subcell resolution. It is not single-point reduced integration of an entire Hex8 element.

For the unit cell at $E=1$, the independent CPU tensor-rule calculation gives:

Poisson ratio	$ K_{\text{composite}} - K_{\text{Gauss}} _F / K_{\text{Gauss}} _F$ (%)	Composite near-zero / positive modes
0.20	0.548361476	6 / 18
0.30	0.516454157	6 / 18
0.45	0.416469489	6 / 18

Here K_{Gauss} is the exact $2 \times 2 \times 2$ Gauss matrix and the eigenvalue tolerance is again $1e-10 \ ||K||_2$. These differences agree with the retained source replay within $1e-12$ in relative Frobenius discrepancy. A literal one-point negative control has eighteen near-zero eigenvalues, but that pathology is not attributed to the pinned composite-rule release. The finding is a mismatch with this study's exact-Gauss operator contract. It does not by itself invalidate the audited mechanics or the conclusions of the original paper. The pinned Futhark constants (c3cd2b00) match exact Gauss to the recorded roundoff tolerance.

Sources: [protocol and pre-execution source observation](#) (`midpoint_points_per_subcell_axis=10`), [retained CPU source replay](#) and [new independent calculation](#). The legacy protocol and raw replay remain unchanged. The generated compact table is [quadrature-checks.tex](#).

Exact counter quantities and profiling boundary

Session J attempt 07 profiled the third launch after skipping two matching launches for each of five FP64 kernels on one RTX 4090 VM and a $128 \times 64 \times 64$ grid (524288 elements). This is one selected launch per kernel; each selected launch was replayed in twelve profiler passes, not twelve independent observations. The input vector is seeded normal (`numpy.default_rng(7)`), and element moduli are $1e-9 + (1-1e-9) \cdot \rho^3$ for seeded uniform ρ in $[0, 1)$.

The three counted metrics are:

```
smsp__sass_thread_inst_executed_op_dadd_pred_on.sum
smsp__sass_thread_inst_executed_op_dfma_pred_on.sum
smsp__sass_thread_inst_executed_op_dmul_pred_on.sum
```

Their sum counts **executed, predicate-enabled FP64 thread instructions**. Each DFMA contributes one instruction here; this is not the roofline's FLOP count, where an FMA contributes two operations. Static SASS instruction sites describe compiled code positions and are a separate quantity. Launched threads, active lanes and element count are also distinct.

Kernel	Executed FP64 thread-instruction total	Per element	FP64 pipe active-cycle peak (%)	DRAM elapsed-cycle peak (%)	Profiled kernel duration (ns)
dense8	314572800	600	93.48	5.19	594688
plain modal8	201326592	384	83.00	7.21	429152
mul-add modal8	125829120	240	88.01	12.14	254304
fused-ai	314572800	600	97.47	5.42	569888
node	402653184	768	94.74	2.26	769568

All totals are divided by 524288 elements. Eight threads per element applies to dense8 and the two modal variants, but fused-ai launches one thread per element and node owns output nodes (including boundary work). Dividing every total by 8×524288 would impose a fictitious shared thread layout. The corrected dense8/mul-add and fused-ai/mul-add count ratios both remain 2.5.

The utilization and time metrics are exactly:

```
sm__pipe_fp64_cycles_active.avg.pct_of_peak_sustained_active
dram__throughput.avg.pct_of_peak_sustained_elapsed
gpu__time_duration.sum
```

The FP64 percentage normalizes to the unit's active cycles; the DRAM percentage normalizes to elapsed cycles. Their percentages therefore have different denominators, and the FP64 value is not an elapsed-time occupancy fraction. The duration is that of the selected profiled kernel, with the raw CSV unit `nsecond`. It excludes wrapper output zeroing, separate mask/projection kernels, host overhead, allocations, transfers and compilation. It must not be substituted for the ordinary complete constrained-product times that include their separate zeroing and projection operations.

The recorded profiler is NVIDIA Nsight Compute CLI **2023.2.2.0, build 33188574**, from `/usr/local/cuda-12.2/bin/ncu`. The commands selected `Speed-OfLight`, `ComputeWorkloadAnalysis`, `MemoryWorkloadAnalysis`, `Occupancy` and `InstructionStats`, together with the requested exact metrics. No clock-control, cache-control or replay-mode override was supplied. The [version-specific CLI documentation](#) specifies defaults `clock-control=base`, `cache-control=all` and `replay-mode=kernel`: base-clock control is attempted, caches are flushed for replay, and kernel memory is saved/restored as required. This is a statement of the recorded command and documented defaults. The retained run does not independently establish successful clock locking, thermal/power stability or a profiler configuration/environment override audit. We do not invent that metadata.

Source: [job/version/command receipt](#), [target](#), the five raw CSVs in `results/session-j-rtx4090vm-20260926-attempt07/remote/ncu/`, and [regenerator](#). `paper/session_j_numbers.json` schema 2 identifies the exact metrics and units, keeps per-metric totals, and records actual launch-thread and replay counts.

Post-hoc accounting and bounded interpretation

For kernel k , define N_k as the count above and U_k as the active-cycle FP64 pipe fraction. The descriptive ratio is $(N_k/U_k)/(N_{muladd}/U_{muladd})$. Its maximum relative discrepancy from the same-launch profiled-duration ratio is 1.768% (approximately 1.8%). Because the utilization denominator already contains active execution-cycle information from that launch, dividing counts by utilization partly recovers cycle accounting. This is a post-hoc consistency result, not an independent prediction of runtime, a held-out validation or an explanation of every timing component.

H15's utilization subcondition passes. Its static-count-ratio subcondition fails: the dense8/mul-add executed ratio 2.5 differs from 78/34 by 8.974%, beyond the declared 5% threshold. This failed declaration remains visible. The smaller plain/mul-add discrepancy (4.615%) does not rescue the joint hypothesis.

High FP64 pipe activity, low DRAM throughput and fewer executed FP64 instructions support FP64 throughput as a dominant constraint for these profiled workloads. They do not independently exclude cache behavior, latency or atomic costs, nor establish a bottleneck on the unprofiled A100, H100 or RTX 3090. Recommendations outside this RTX 4090 workload must instead be bounded by their own measured product and solver endpoints.

Hardware basis and idealized traffic of the roofline

The original declared roofline used the A100 80 GB **SXM** specification: 9.7 non-tensor FP64 TFLOP/s and 2039 GB/s. Its original FP64 ridge 4.7572 FLOP/B is retained and labeled. For the measured A100 80 GB **PCIe**, the SKU-specific calculation uses 9.7 FP64 TFLOP/s, 19.5 FP32 TFLOP/s and 1935 GB/s from [NVIDIA's A100 specification table](#). The corresponding ridges are 5.0129 FLOP/B (FP64) and 10.0775 FLOP/B (FP32). The ideal dense/parity ratio remains 1.0 in both precisions. The added row is an added SKU calculation; it is not relabeled as a prospective prediction. Both hardware bases and their status are emitted by [roofline.py](#) to [roofline.json](#) and [roofline.md](#).

The 48 values per element count one gather of 24 displacement values and one scatter of 24 results. This is idealized traffic accounting, not measured DRAM bytes or a universal traffic lower bound. Shared-node reuse and cache hits can reduce DRAM loads; atomic read/modify/write transactions, modulus reads and constraint masks add traffic. The model excludes these effects and cannot establish their measured share of runtime.

Regeneration and verification record

Executed on the local CPU on 2026-09-27:

```
.venv/bin/python analysis/supplement_algebra.py
.venv/bin/python analysis/session_j_counters.py
.venv/bin/python analysis/roofline.py
.venv/bin/python -m pytest tests/test_kernels_cpu.py experiments/session_e/test_session_e.py -q
```

The algebra generator passed both geometry checks and all three quadrature cases. The existing tests passed **16/16**. The butterfly hard corpus emits six expected overflow/invalid-operation warnings while testing extreme inputs; all comparisons pass. The tests include FP64/FP32 lane emulation against assembled products, shuffle negative control, source equality for the butterfly variants, and the local bitwise corpus. This does not replace GPU execution or GPU accuracy replay. Counter totals were read directly from all five retained CSVs, then normalized by the protocol element count; H15's failed count condition remains failed. The parser additionally checks one data row per selected kernel and the raw units `inst`, `%` and `nsecond`. Direct checks of the generated JSON against the five raw totals passed; both dense8/mul-add and fused-ai/mul-add ratios equal 2.5. Both original and reconstructed stiffness matrices have six near-zero and eighteen positive eigenvalues under the stated tolerance. The added PCIe and original SXM roofline records retain their distinct bandwidths and both give ratio 1.0.

Physical problems and solver accuracy

This supplement specifies the implemented problems, including differences between experiment families. Paths refer to the accompanying artifact.

Constitutive and discretization contract

Small-strain isotropic elasticity uses $\sigma = 2 \mu \epsilon + \lambda \text{tr}(\epsilon) \mathbf{I}$, with $\mu = E/[2(1+\nu)]$ and $\lambda = E \nu/[(1+\nu)(1-2\nu)]$. The reference element uses $E=1$, $\nu=0.3$, trilinear Hex8 interpolation and exact tensor $2 \times 2 \times 2$ Gauss integration. Each element has a spatially constant scalar

modulus $E_e = E_{\min} + (1 - E_{\min}) \rho_e^3$; this scalar multiplies its reference stiffness. The parity exactness claim does not cover distorted cells, anisotropic constitutive laws, or a tensor varying within an element.

Family	Element geometry/material	Densities and boundary/loading contract
Sessions A–E, base cantilevers	Cubes of side $1/q$, grid $2q \times q \times q$; $E_{\min}=1e-6$, except explicitly labeled Session C contrast cases $1e-9$	Clamp all displacement components on $x=0$. On the index line $x=n_x, z=0$ (physical $x=2, z=0$), apply negative z load with interior nodal weights $1/ny$ and endpoint weights $1/(2ny)$, total force -1 . Retained optimized fields at $q=64/96$; uniform control $\rho=0.12$; smoke control $q=4$ and $\rho=0.12$.
Session I rediscretized scale	<code>make_inputs.upsample</code> repeats each density and preserves the base <code>ke</code> byte for byte; the effective fine-element stiffness scale remains that of the $q=64$ or $q=96$ base, with coarse matrices scaled by 2^{level} ; $E_{\min}=1e-6$	Same clamp and total unit line-load rule on the enlarged grid. These are repeated coefficient fields, not new optimized designs. In particular, no extra $1/\text{factor}$ stiffness scaling is applied during upsampling.
Sessions F, G, H and I Galerkin branch	Donor <code>KE_UNIT_3D</code> , the unit-cube matrix (not rescaled using the preset's domain coordinates); $E_{\min}=1e-9$, penalty 3, $\nu=0.3$	Preset coordinates select supports/load nodes; force vectors and dimensions follow the pinned donor presets below. F integration: $\rho=0.5$ or uniform random ρ in $[0.3, 1]$, NumPy seed 20260926. F sizing and I Galerkin: $\rho=0.5$. G/H start from the preset volume fraction and use a density filter of radius 1.5 element spacings, move 0.15, top88 update without continuation/projection.
Session J profiling	Unit-cube <code>ke_unit_3d.npy</code> ; $E_{\min}=1e-9$, penalty 3	NumPy <code>default_rng(7)</code> : standard-normal displacement vector followed by uniform $[0, 1]$ densities. This is a product workload; no physical solve or load case is inferred from it.

The base force/mask are reconstructed by `experiments/session_i/make_inputs.py:regenerate`; factor-one equality is checked in `test_session_i.py`. The uniform density is read directly from `experiments/session_e/inputs/uniform-q64.npz`. Material and stopping parameters are in each frozen protocol.

G/H case (preset)	Grid	Preset domain	Initial/target volume	Supports and force
cantilever-216k (<code>cantilever_gpu_large</code>)	$120 \times 60 \times 30$	$2 \times 1 \times 0.5$	0.30	Clamp left face; $(0, -1, 0)$ point force nearest $(2, 0.5, 0.25)$
cantilever-512k (<code>cantilever_gpu_xlarge</code>)	$160 \times 80 \times 40$	$2 \times 1 \times 0.5$	0.30	Same normalized location and unit force
mbb-514k (<code>mbb_gpu_xlarge</code>)	$210 \times 70 \times 35$	$3 \times 1 \times 0.5$	0.50	Left-face <code>pin_x</code> fixes only u_x ; point <code>pin_y</code> fixes only u_y nearest $(3, 0, 0.25)$; $(0, -1, 0)$ force nearest $(0, 1, 0.25)$. No u_z constraint is added by this preset.

G/H case (preset)	Grid	Preset domain	Initial/target volume	Supports and force
torsion-499k (torsion_gpu_500k)	165×55×55	3×1×1	0.25	Clamp left face; forces (0,−0.5,+0.5) at (3,1,0.5) and (0,+0.5,−0.5) at (3,0,0.5)

These are the exact pinned `presets.py` and `bc_generator.py` conventions, including nearest-node snapping via `argmin` and removal of loads on constrained DOFs. Domain coordinates specify that snapping; the unit-element stiffness in the donor runner is a separate convention and is not silently replaced by physical-domain-scaled stiffness.

F integration uses the same cantilever domain/support/load convention, with the 80×40×20 `cantilever_gpu_medium` grid and an overridden 37×23×17 odd grid. F sizing uses 252×126×63 (2M) and 342×171×86 (5M) first, with later declared sizes 432×216×108, 544×272×136 and 640×320×160 subject to the stop-on-failure ladder. I's Galerkin branch uses 200×100×50 and 252×126×63. All of these retain the same unit-cube stiffness convention and point force; they do not use the rediscretized family's line load. These are actual implementation conventions, not a claim that the dimensioned preset boxes set each stiffness length scale.

Residual and energy definitions

Let M be the diagonal free-DOF mask and $A=M^T K M + (I-M)$. The full-vector residual is $\|A^{-1/2}(u-f)\|_2 / \|f\|_2$. Saved states must be finite and exactly zero at constrained DOFs; f is also zero there, so this equals the free-space norm. For the rediscretized CPU/C-reference gate, energy-work discrepancy is $|\sum_e E_e u_e^T K_0 u_e / (f^T u) - 1|$; the numerator is twice the strain energy, with no extra 1/2. These checks compare the same physical units. `analysis/supplement_accuracy.py` regenerates maxima, counts and source paths in `accuracy.json` and `accuracy.md`. Their E/I maxima cover all retained in-run independent checks, not just saved-state replay: 440 states for Session E across two hosts, 36 for RTX scale and 96 for A100/H100 scale. Separate retained local Session E replay covers the first repetition for each case/arm (44 states per host, 88 total); its largest residual is 4.35296445e-9. Session I retains in-run CPU/C-reference checks and solution hashes, without a separate saved-state replay being claimed here.

Experiment/endpoint	Requested stop	Independent acceptance	Coverage/exception
A–C solves to tolerance; D/E rediscretized multigrid	true relative residual 1e-8	CPU reference residual $\leq 1e-7$; energy-work $\leq 1e-6$; finite state and exact fixed zeros	Accuracy gates are separate from timing; fixed-work runs intentionally do not stop by tolerance.
I rediscretized multigrid	true relative residual 1e-8	C-reference residual $\leq 1e-7$; energy-work $\leq 1e-6$; exact fixed zeros	Checks executed during retained GPU runs; reported maxima are from those records.
F Galerkin integration	1e-10, max 300 iterations	stock-operator true residual $\leq 1e-10$ and solution relative L2 agreement $\leq 1e-8$	Product agreement $\leq 1e-12$. Mixed random-density cases on A100/H100 hit cap; integration gate failed.
F Galerkin sizing	1e-8, max 40 iterations	native convergence flag/iteration count only in sizing records; no separate stock residual replay	A successful allocation/sizing run is not proof that the capped solve met tolerance.

Experiment/endpoint	Requested stop	Independent acceptance	Coverage/exception
G/H Galerkin optimization	donor default CG tolerance 1e-5, max 1000, warm starts; automatic method resolves to PCG for these FP64/FP32 smoother schedules	Recorded linear-solve gate uses the iteration-cap proxy (<code>last_cg_iters < c_g_maxiter</code>); final compliance relative difference $\leq 1e-3$ against stock on repetition 1, final physical volume within $1e-3$ of target on all fixed trajectories	Donor PCG tests recursive residuals (plus its initial true residual), rather than recomputing true residual every iteration. G/H outputs do not carry the E/I independent C-reference residual/energy check. Capped designs are not converged optimizations.
I Galerkin branch	1e-8, max 1000 for preliminary solves	retained donor convergence records; fixed-work endpoint uses 50 iterations	Do not infer the rediscretized branch's independent accuracy gate for this branch.
J product counters	no iterative stop	no solver residual endpoint	One profiled kernel launch, not a solve.

A–C test the Jacobi/iterative-refinement outer solution's true residual at their declared checks (every 50 PCG iterations); they are not instances of the D/E/I flexible-CG recurrence below. The donor G/H solver also retries failed warm-start solves from zero. Its cap proxy and compliance/volume comparisons are reported as those checks, without converting them into a broader independently verified residual claim.

Flexible CG and product accounting

The rediscretized hierarchy halves all three extents while $n_x > 16$ and every extent is even. It stops early if another halving would be nonintegral: the largest q96x3 hierarchy ends at $18 \times 9 \times 9$, for example. Noncoarsest operators are matrix-free, with averaged child moduli and $K0_coarse = 2^{level_K0_fine}$; the coarsest constrained matrix is assembled and factored by dense Cholesky.

`src/voxel_mg.py:fcg` implements FP64 Polak–Ribière flexible CG, including when the preconditioner computes its V-cycle in FP32. Start with $x=0$, $r=b$, $z=B(r)$, $p=z$, $\delta = r^T z$. Each iteration computes:

```

q = A p
alpha = delta / (p^T q)
x = x + alpha p
r_new = r - alpha q
if a requested true-residual check is due:
    r_new = b - A x
    stop if ||r_new|| / ||b|| <= tolerance
z_new = B(r_new)
beta = r_new^T (z_new - z) / delta
p = z_new + beta p
delta = r_new^T z_new
r, z = r_new, z_new

```

The checked residual replaces the recursive residual. The routine defaults to `check_every=5`, but D/E/I frozen protocols pass `check_every=1`: a successful to-tolerance iteration therefore costs two outer products (search direction and true residual). Fixed-work mode skips intermediate replacement and uses K direction products plus one final true-residual product, $K+1$ total. A capped to-tolerance exit executes the final check as well: with `check_every=1`, a cap of K iterations costs $2K+1$ outer products if no early return succeeds. This recurrence describes the rediscretized solver; G/H retain the separate donor solver and its tolerance.

Damping estimator and limits

spectral_omegas estimates the spectral radius of constrained/free-space $D^{-1} A$ on each noncoarsest level. It initializes a NumPy standard-normal vector, masks fixed DOFs and normalizes it. For 30 iterations it computes $w = D^{-1} A v$, masks fixed DOFs, sets $\lambda = \|w\|_2$ and $v = w/\lambda$. Then $\omega = 4/(3\lambda)$. The random generator is restarted on each level with the same seed, not advanced continuously across levels. Default seed is 20260923; runners pass their protocol order_seed (D/E: 2026092320; I: 2026092621). The reference FP64 hierarchy uses node-owned products; every arm shares its estimated weights and the same host-computed coarsest factor, with that factor cast to the arm's preconditioner precision on construction.

Thirty power steps are an estimate, not a certified upper eigenvalue bound. Empirical evidence consists of CPU smoother/transfer tests and the retained successful residual/energy checks for the tested coefficient fields. A fixed 0.6 damping failed heterogeneous CPU cases. Neither the estimator nor those tests establish universal smoother stability or mixed-precision convergence. The exact rational factorization certificates in the separate pinned-release audit concern different ideal operators and must not be attributed to this power estimate.

Verification on 2026-09-27: `.venv/bin/python -m pytest tests/test_mg_cpu.py -q` passed all 10 existing CPU checks, including transpose transfers, uniform-modulus Galerkin equality, tested V-cycle symmetry, FP32-preconditioned convergence and fixed-work product accounting. The five existing Session I checks also passed (`.venv/bin/python -m pytest experiments/session_i/test_session_i.py -q`), including input regeneration and the CPU dry-run scale path. Input inspection independently confirmed base ke against exact Gauss cells of side $1/q$ to about $3.1e-16$ relative Frobenius error, and exactly reproduced every base mask and force using regenerate. These are CPU checks and source/record inspection, not new GPU validation.

Retained accuracy results

Generated by `python analysis/supplement_accuracy.py`. This inspects retained independent-check records; it is not a new GPU experiment or fresh state replay.

Experiment	In-run checked states	Requested	Independent acceptance	Observed residual maximum	Energy-work maximum (limit 1e-6)
Session E, two RTX 4090 hosts	440	1e-08	1e-07	4.35301436e-09	7.94719845e-12
Session I, RTX 4090	36	1e-08	1e-07	4.43192315e-09	3.06175085e-11
Session I, A100 and H100	96	1e-08	1e-07	7.44563488e-09	6.84184931e-11

Source records:

- `results/session-e-host1-20260925/remote/run-01/result.json`
- `results/session-e-host2-20260925/remote/run-01/result.json`
- `results/session-i-rtx4090-20260926/remote/run-01/result.json`
- `results/session-i-a100-20260926/remote/run-01/result.json`
- `results/session-i-h100-20260926/remote/run-01/result.json`

Separate retained local CPU replay of saved Session E states (first repetition only):

- `results/session-e-host1-20260925/local-verification.json`: 44 states, maximum residual $4.35290392e-09$; all passed.

- `results/session-e-host2-20260925/local-verification.json`: 44 states, maximum residual 4.35296445e-09; all passed.

Session I records CPU/C-reference checks during each GPU run and solution hashes; this summary does not claim a separate saved-state replay for that family.

Timing, optimization outcomes and measured scale

This supplement documents optimization outcomes, solver performance and measured capacity. Its [generated tables](#), [exact JSON](#) and CSV files are regenerated from retained evidence by `python3 analysis/supplement_outcomes.py`. This is numerical regeneration on the CPU, not fresh GPU execution. Every optimization, scale-solve and Galerkin record includes its raw file and JSON pointer; input hashes are in the JSON. Historical fields named `complete` or `complete_optimization` are retained as evidence identifiers and do not establish design convergence.

Experiment manifest and timing boundaries

The [manifest CSV](#) and generated hardware table list GPU SKU/capacity, CPU, driver, CUDA runtime, CuPy, compiler, host/allocation count, cases, arms, repetitions, warmup and endpoints for 20 measured allocations across A–J, synthetic replication and libCEED. An unavailable entry is NR/null. NVRTC is the CUDA just-in-time compiler; a Python build's GCC version does not identify the C++ reference compiler. The native libCEED harness retained `nvcc/GCC` version commands; reference-compiler versions were not separately retained in the CuPy session records. CUDA driver capability, locally installed runtime and CuPy-linked runtime are different fields. Where the installation record distinguishes them, its full value is preserved. Clock telemetry is a snapshot; no common fixed-clock policy is claimed. Manifest rows identify the metadata files examined. The only missing CPU model is Session A: its manual rental/preflight records 32 CPUs and 188 GB RAM, but no CPU model; the retained installation and result records do not supply one.

The synthetic FP64 product study directly measured one RTX 4090 and an independent RTX 3090, with plain parity, dense8 and the three donor products. It used five warmup calls, eleven paired rounds and twenty products per timed batch. Session A used one RTX 4090 for physical-state products/Jacobi-PCG/refinement, B one **A100 80GB PCIe**, and C another RTX 4090 for refinement and modulus-floor sensitivity. A/B had three tolerance repetitions; C had two. Their product measurements used eleven rounds of twenty products, and fixed-work solves ten paired rounds. Session D used one RTX 4090 for rediscritized multigrid, five tolerance repetitions and ten fixed-work rounds. Session E used two RTX 4090 hosts, eleven product rounds of twenty products, five tolerance repetitions and ten fixed-work rounds. Full cases and kernel variants are preserved in the manifest's structured columns and the named protocols.

Session F's feasibility cards were RTX 4090, A100 80GB PCIe and **H100 NVL (95830 MiB reported physical memory)**. The Session I scale H100 was **H100 80GB HBM3 (81559 MiB)**. They are different hardware. No measured A100 PCIe result is relabeled as an A100 SXM measurement; the historical SXM specification used in the roofline calculation is a separate model input.

Session G ran three paired 30-step trajectories per case/schedule on each of two RTX 4090 hosts. H ran three paired **100-step** trajectories on each of two other RTX 4090 hosts. Each warmed one step per case/schedule/arm. Each also retained one additional trajectory per arm/schedule for the 216k cantilever, stopping on design change or the 150-step cap. There are three arms (stock gather/DGEMM/scatter, fused-ai FP64, parity mul-add) and two four-level Galerkin schedules (FP64 and mixed). Session I used one allocation per GPU type: RTX 4090, A100 80GB PCIe and H100 80GB HBM3. It used five rounds of ten products, two tolerance solves per arm and three paired fixed-work rounds, plus one instrumented profile per selected arm/case. Its W2a Galerkin check used two blocks with three fixed 50-iteration solves per block/arm/schedule and one separate tolerance attempt in block zero.

Session J used one RTX 4090 VM for the successful counter collection, five FP64 kernels at 128×64×64 elements, and one selected launch per kernel after skipping two matching launches. Nsight Compute 2023.2.2.0 replayed the selected launch in twelve metric passes; these are not twelve independent timing samples. Profiler kernel duration excludes wrapper zeroing/projection and host overhead. The installation logs identify cudart/NVRTC 12.2.140; no separate loaded-runtime version query

is retained. [Counter methods](#) retain clock/cache/replay policy and distinguish unavailable achieved-clock metadata. The libCEED native CUDA/C harness used one RTX 4090 allocation, FP64, six timed shape/pattern combinations from $64 \times 32 \times 32$ to $192 \times 96 \times 96$, five seconds of warmup and twelve randomized blocks of 256 actions. Each action zeroes output, applies the operator and identity rows, then synchronizes; host steady-clock time is primary. Setup, allocations, conversion, transfers and compilation are excluded. Byte-verified metadata extracted from the deposited files archive records the Ryzen 5 5600X CPU, driver 595.58.03, 24564 MiB GPU memory, nvcc 12.6.85, GCC 11.4.0 and CUDA 12.6 SDK probe. Extraction member names and hashes are in `provenance/supplement/manifest-metadata/extractions.json`; the two original synthetic product hosts' rental/CuPy metadata was recovered by the same method. See [comparator provenance](#).

Endpoint	Included in its timer	Excluded or separately recorded
Synthetic and A–C complete constrained product	Output zeroing where needed, mask/projection, launch, assembly/atomic scatter and constrained identity restoration	Input generation, host/device transfer, setup/JIT, validation; resident buffers reused
E/I isolated raw product	Output zeroing and operator action; CUDA-event elapsed time	Mask/projection is outside the raw product call; constrained entries of the probe were zeroed; transfers/setup/JIT/replay excluded
D/E/I fixed-work solve	Synchronized solver wall time, vector allocations performed by the solver, prescribed Krylov iterations, all V-cycle/outer actions, vector operations and reductions	Hierarchy construction, coarse factorization, spectral estimate, compilation warmup, host transfer and independent physical check
D/E/I solve to tolerance	Same solver boundary, including each true-residual recomputation/replacement and stopping test	Same setup/replay exclusions; not the same product count as a fixed-work run
G/H total optimization wall	Solver construction, initial allocations/transfers, every solve, density-dependent hierarchy setup, filtering, sensitivities, OC update and synchronization	One-step compilation warmup precedes timed trajectories; state saving and final cleanup occur after timer stops
F capacity probe	Cold constructor/setup and a separately timed capped solve; initial compilation may occur	It is not a warm solve to tolerance
I W2a	<code>setup_seconds times gmng . setup (E)</code> ; fixed-work time brackets PCG and its synchronized allocations/actions	BC generation, operator/GMG construction precede setup timer; setup is not the full construction cost; GPU memory sample is after setup

The G/H `gmng_setup_ms` sum is the repeated density-dependent setup time. Initial solver/hierarchy construction is in the total wall time but not isolated by a retained timer; it cannot be recovered by subtracting CG/setup/sensitivity totals because the remainder also includes filtering, OC, transfers and other work. Thus the table reports that construction component as NR instead of inventing a split. For H, density filtering and OC are GPU resident; G performs these on the CPU and includes associated transfers. Warmup removes many JIT costs but does not prove that every later call is compilation-free.

Session I shared setup includes transfers, construction of reference hierarchy, host dense coarsest assembly/Cholesky and 30-step spectral estimation. Per-arm setup is the recorded incremental construction time with possible reuse of compilation/cache state, not six independent cold costs. All six arms are resident in the measured harness before solves. Its reported fitting size and OOM are therefore implementation/harness capacity observations, not isolated per-arm optimal capacity limits.

Uncertainty and decisions

Session E computes a paired ratio for each **round**: numerator time divided by candidate time. The statistic is the median of these ratios, not the ratio of medians. Its one-sided 95% lower bound is the 5th percentile of 10,000 bootstrap sample medians, resampling the round ratios with replacement using NumPy `default_rng(order_seed)` (2026092320); sample size equals the number of paired rounds. Twenty products within one round are repeated work in a batch, not twenty independent observations. See `experiments/session_e/analysis.py:bootstrap_lower` and the frozen protocol's H8/H9 rules. H8 requires median ≥ 1.05 and lower

bound ≥ 1.02 against the fastest declared donor on both primary cases, with all rounds and checks present. H9 requires its separately declared all-FP64 fixed-work thresholds. The synthetic study uses its own bootstrap seed (20260924) and declared thresholds; A–D use their frozen analysis/protocol rules.

A narrow interval such as 1.259 with lower bound 1.258 describes within-session round variability conditional on the measured host/configuration. It does not quantify uncertainty over rental hosts, GPU populations, arbitrary designs or the choice of implementation. The two-host E replication is a separate outcome, not a bootstrap population sample.

G/H use **median-threshold decisions**, not one-sided bootstrap tests. For each primary all-FP64 case on a host, H11/H11b requires median paired wall ratio ≥ 1.10 , at least two completed repetitions, total Krylov counts matched within 3% in every pair, and the compliance/volume/solve gate. Replication requires both hosts to pass. The generated paired table prints all three observations, including variability and unfavorable results. Mixed-schedule H12/H12b comparisons are descriptive medians with a declared expectation below 1.10; they are not confidence-bound decisions. Single capped trajectories have no repeated-sampling uncertainty estimate. Session I H13/H14 also use declared median thresholds on three fixed-work rounds; two tolerance times support a descriptive ordering among the tested routes, not a statistical universal winner.

Optimization outcomes and validation coverage

The [outcome CSV](#) contains all 288 fixed-length trajectories (four hosts $\times$ four cases $\times$ two schedules $\times$ three arms $\times$ three repetitions), 24 capped trajectories and two Session G auxiliary diagnostics. No diagnostic, cap or unfavorable ratio is discarded. It records step count, exit reason, total Krylov count, wall time, CG time, hierarchy re-setup, sensitivity/diagonal times, final logged compliance, volume and design change. The readable table gives separately computed medians per host/case/schedule/arm, with $n=3$ for fixed runs and $n=1$ for capped runs. Exact per-repetition values and sources remain in CSV/JSON.

The donor log's final compliance is evaluated at the density **before the last OC update**. Its final volume and design change refer to the physical density/new design after that update. These are the retained endpoint conventions, not a new equilibrium solve on the saved final density. Design change is the infinity norm of the unfiltered design-variable update $\max(\text{abs}(x_{\text{new}} - x))$; the cap criterion implemented in the frozen runner is $\text{change} \leq 0.01$. All 150-step trajectories exited at the iteration cap. For example, H host 2 FP64 parity ends at change **0.1484386706** (G host 1: **0.1484407262**), far above 0.01. Successful linear solves and matching compliance do not imply that the design has converged. These are capped end-to-end trajectories; no time-to-converged-design claim is made.

The fixed-run agreement gate compares each nonstock arm's repetition-1 final logged compliance to stock, using $\text{abs}(C_{\text{arm}} - C_{\text{stock}}) / \text{abs}(C_{\text{stock}}) \leq 1e-3$. Every fixed run is checked for final physical volume within absolute $1e-3$ of the preset volume fraction and no solve whose iteration count reaches the 1000-iteration cap. This is compliance/volume agreement under that coverage; it is not density-field equivalence. The current analysis does not compute a density-field norm, maximum per-step compliance difference, or a same-state independent residual for every optimization solve. Stored design arrays cover repetition 1 and the capped trajectories; repetitions 2/3 have scalar histories, not stored fields. The frozen protocol's intention to report field norms must not be confused with completed verification. The exposed equivalence compatibility field has precisely this narrower meaning.

For H's **three primary 100-step all-FP64 cases**, median fused-ai/parity wall ratios span **1.080900–1.181485** over both hosts. Adding the 216k cantilever gives **1.023073–1.181485**. The fourth-case ratios are **1.025590 / 1.023073** on hosts 1/2. Its single capped FP64 trajectory gives **1.020899 / 1.021647**. Mixed runs and G's CPU-optimizer pipeline are separate populations: see the complete paired table, which includes H host 1's capped mixed ratio **0.850511**. None of these selected ranges describes every case, schedule and trajectory length.

The earlier “about 25 s instead of 66 s” values are the medians of the six capped runs across three arms and two schedules, **H host 2: 24.939278861 s; G host 1: 66.342864434 s**. They illustrate different pipelines on different rental hosts. They are not paired observations and do not isolate the causal effect of moving the optimizer to the GPU. Within-host, within-schedule arm comparisons remain the controlled performance evidence.

Practical kernel map and scope

“Fastest” below means fastest among the retained tested implementations, sizes, precision schedules and endpoint definitions. All statements are limited to axis-aligned isotropic Hex8 with the tested scalar SIMP modulus fields; enlarged coefficients are nearest-neighbor copies of the q64/q96 design family. This is neither arbitrary-contrast validation nor a general mixed-precision convergence result.

Hardware/evidence	Isolated FP64 product	Finest/outer choice in all-FP64 rediscretized solve	Fastest observed mixed tolerance route
RTX 3090 synthetic replication; 0.52/1.77 M elements	Plain parity directly measured	Mul-add/multigrid unmeasured; any suggestion is an extrapolation from 4090	Unmeasured
RTX 4090 E, two hosts; 0.52/1.77 M elements, and I, one host; 1.77–14.16 M	Mul-add directly measured, including comparison with plain parity	Mul-add, fixed work; I 43.0 M total DOFs gives fused-ai/parity 1.352031 , all-FP64	FP32 node V-cycle + FP64 parity mul-add outer, directly measured
A100 80GB PCIe B/I; up to 47.78 M elements in I	Node fastest among tested I FP64 products; fused-ai is not the fastest A100 product comparator	Node; I parity/node fixed-work ratio 1.189658–1.200122	At q96x1, fused-ai outer 0.316095714 s , node outer 0.319631294 s ; node outer wins enlarged cases
H100 80GB HBM3 I; 1.77–47.78 M elements	Node in tested I product timings	Node; parity/node 1.179589 at q96x1 and 1.091656–1.104500 in enlarged primary cases	FP32 node V-cycle + FP64 node outer in tested cases

The A100 q96x1 difference is about 1.1% and uses only two tolerance repetitions per route on one host. It is a small observed advantage for fused-ai outer, not evidence that node always wins, nor proof of statistical equality. Product, all-FP64 fixed-work and mixed time-to-tolerance rankings must not be substituted for one another. The H100 1.09–1.10 range applies only to the enlarged primary cases; the continuity case is about 1.18.

Measured capacity and allocation semantics

The [scale table](#) reports every successful hardware/case/arm pair: exact elements, total/free DOFs, hierarchy depth, iteration counts, two warm solve times in the CSV, median in the readable table, shared and incremental setup. Rediscretized multigrid means matrix-free rediscretization on **noncoarsest** levels and a small assembled/factored coarsest operator solved by dense Cholesky. It does not mean no coarse operator is stored.

The largest successful RTX 4090 case was q96x2: **14,155,776 elements; 43,022,595 total DOFs; 42,910,848 free DOFs**. Its fastest observed warm tolerance route was FP32 node V-cycle / FP64 parity outer, median **2.116005612 s**. This endpoint differs from the approximately 1.35× **fixed-work all-FP64** speed ratio above. The q96x3 construction failed with OOM in the six-arm resident configuration. A100 and H100 completed q96x3: **47,775,744 elements; 144,574,851 total DOFs; 144,324,288 free DOFs**, with fastest mixed medians **4.536915969 s** and **3.977970041 s**, respectively. Counts are calculated from retained grid dimensions, not rounded protocol descriptions.

Nearest-neighbor upsampling of the retained design fields demonstrates capacity and solver behavior for those fields. There is no independently optimized design at each resolution, general mesh-independent convergence result, or parallel strong/weak-scaling experiment. Different q64/q96 base fields also preclude treating the entire ladder as one identical-coefficient refinement sequence.

Session F’s pinned Galerkin implementation allocated about **46.3 GiB device-used / 46.3 GiB pool-reserved** for its successful 2.000376 M-element setup on A100/H100 NVL, or about **23 GiB per million elements**. This is a property of that implementation, construction arrays and allocator behavior, not an intrinsic requirement of all Galerkin methods. Device-used is total-minus-free CUDA memory; pool-reserved includes cached blocks and does not equal live arrays. Separate live-array totals and transient construction peaks were not measured. The allocation-failure message identifies a requested temporary allocation, not a successfully measured peak. The 2

M setup and 40-iteration solve succeeded as operations but did **not** reach tolerance. The 5.029452 M setup failed on both large cards; RTX 4090 failed already at 2 M. Approximately 3 M on an 80 GB card is only an estimate between observed trials, not a measured maximum.

W2a records every 1 M/2 M arm/schedule/block, including setup, device-used memory, 50-step fixed-work timing and the first tolerance attempt. The 1000-iteration cap failures remain labeled failures to converge; capacity or fixed-work timing is not converted into time to tolerance. The tables explicitly distinguish the feasibility H100 NVL from scale H100 HBM3 and the different setup timing boundaries.

Comparator denominators and session-specific ratios

The product comparison reports **fused-ai time / plain-parity time**. The practical kernel-map figure uses speed relative to the fastest tested donor product, which is node on A100; these denominators differ and are labeled. The RTX 4090 Nsight Compute examination in [counter methods](#) provides the retained hardware-counter evidence for the throughput interpretation.

For Session E host 1 q96, the approximately $0.752 / 0.326 = 2.31$ gain combines an FP32 node V-cycle **and** replacement of fused-ai outer by parity mul-add. Keeping fused-ai outer, $0.752 / 0.369 = 2.04$ is the corresponding V-cycle precision/configuration change. The generator now supplies FpThirtyTwoVcycleG ainQnineSix as well as the combined macro.

The FP64 parity-PCG **20.3401218606 s** value belongs to Session C q96/Emin=1e-6 (median of two repetitions); **19.9638886160 s** belongs to Session A q96 (median of three repetitions). Different RTX 4090 allocations and session statistics explain the two values; they are not two representations of one measurement.

Regeneration

```
.venv/bin/python analysis/session_g_tables.py results/session-g-host1-20260926 results/session-g-host2-20260926
.venv/bin/python analysis/session_h_tables.py results/session-h-host1-20260926 results/session-h-host2-20260926
.venv/bin/python analysis/session_i_tables.py \
  results/session-i-rtx4090-20260926 results/session-i-a100-20260926 results/session-i-h100-20260926
.venv/bin/python analysis/supplement_outcomes.py
.venv/bin/python paper/manuscript/make_figures.py
```

These commands preserve all raw results, saved states, frozen code and protocols. Generated summaries retain every case and expose capped exits, setup components and statistical coverage rather than silently strengthening the original observations.

Comparator provenance, release audit and claim evidence

Evidence check date: 2026-09-27. This supplement documents comparator provenance, pinned-release audits and claim evidence. Paths are relative to the artifact repository unless explicitly described as members of a deposited archive. Historical protocols and results remain unchanged. Numerical values are summarized from retained records, not from new GPU runs.

Contribution and closest comparisons

The engineering contribution is an exact low-order Hex8 parity implementation, comparison of its floating-point butterfly forms, and controlled measurement of where its product advantage survives in solver and optimization contexts. The tensor sign basis, reflection decomposition and Hadamard butterfly are established mathematics.

The signed multiply-add implementation is an FP64 adaptation of Tri Dao's `fast-hadamard-transform`, commit `e7706faf8d1c3b9f241e36860640ad1dac644ede`, `csrc/fast_hadamard_transform_common.h`; attribution and BSD notice are retained in `experiments/session_e/dependencies/ATTRIBUTION.md` and `DAO-LICENSE`. No claim that all earlier Q1 methods lack a basis change is needed.

Implementation / closest work	Element and operator	Integration / coefficient	Layout and accumulation	Precision and reduction
This parity product	Axis-aligned rectangular Hex8, isotropic elasticity, constant tensor per cell times scalar modulus	Exact $2 \times 2 \times 2$ Gauss law; per-cell scalar SIMP modulus	Eight lanes per element, analytic indices, shared-node atomic scatter	FP64 local parity blocks; three butterflies tested; FP32 counterpart tested separately
Kiran, Gautam and Sharma (2020)	Four-node quadrilaterals ; elasticity and scalar heat examples	Supplied element stiffness; the selected method screen did not establish the integration rule	Unstructured mesh data; element submatrices, warp communication, coloring to prevent conflicting updates	Double precision; triangular symmetry/storage reduction, rather than this Hex8 reflection-block construction
Yang fused release, <code>fused_fp64 / fused_ai_fp64</code>	Cartesian Hex8 elasticity, dense 24×24 element action	Same caller-supplied exact-Gauss matrix and per-element modulus in this comparison	One thread per element; explicit edof / analytic indexing; atomic scatter	Supplied FP64 code paths; no parity basis reduction
Same release, <code>node_fp64</code>	Same physical operator	Same matrix and modulus	One thread per output node, incident-element gather, no scatter atomics	Supplied FP64 instantiation; duplicated incident-element reads and fixed accumulation order
Yang Galerkin stock FP64	Same low-order elasticity family	Caller-supplied fine matrix; assembled Galerkin coarse levels	CuPy gather, DGEMM, bincount scatter in <code>MatrixFreeKff</code>	FP64 stock fine action; donor FP32/BF16 fused paths are different schedules
Tested libCEED configuration	Unit-cube Q1 vector elasticity	Tensor $2 \times 2 \times 2$ Gauss; scalar density per cell	Natural interleaved node layout and element restrictions; CUDA restriction transpose performs assembly	FP64 <code>/gpu/cuda/gen</code> , with <code>/ref</code> and <code>/shared</code> controls; tensor basis action, not an implementation-wide library ranking

The Kiran element class, double precision and colored element assembly are documented in the [author-hosted manuscript](#), selected methods and test-problem sections. The table explicitly leaves unverified details unidentified instead of inferring them from the element name.

Exact Yang source identities and adaptations

The cited versions are [arXiv:2604.18020v1](#) and [arXiv:2604.26441v1](#), both preprints. The first paper emphasizes an FP32 fused framework. That paper's FP32 performance claims do not establish the FP64 performance measured here.

The retained fused donor matches [release commit 3a6e37af137c3f8ba65ef1d298602a6390da6466](#). A read-only source comparison found the following exact byte identities (retained in `provenance/supplement/comparator-identity.json`):

Retained file under experiments/session_e/donor/ src/gpu_fem/	SHA-256	Symbols / status
cuda_operators.py	d36cc55cf762e6f701056aeccc18977f8f4e0914b e278c6971162fca0db7dc6f	fused_matvec_fp64, fused_matvec_ai_fp64, node_matvec_fp64 are supplied in this release. _instantiate generates the latter two from release-owned __T__ templates with double; this substitution is upstream code, not a new investigator adaptation.
cuda_fused_matvec.py	7e9f7002451b1b118243410ad62a3a6f19e9fc5c0 99ad2e68ca6af0575f2aa0f	Supplied FP32/BF16 fused kernels; the FP64 baseline should be attributed to cuda_operators.py.

The Galerkin donor is pinned to fd9ddecd039fd9b7fffb8cf49c58f602318ed2ef in experiments/combined/donor_yang_gmg/PINNED.txt. Its src/gpu_fem/solver_v2.py:MatrixFreeKff.matvec implements the stock FP64 gather/DGEMM/bincount path. experiments/combined/kff_adapters.py:ArmKff replaces the FP64 fine core, including the fine smoother's FP64 calls; it preserves the donor free/full expansion and restriction. Non-FP64 products and diagonal extraction remain donor calls. Thus “unchanged” describes the imported source files, not the complete adapted experiment. The common path-discovery workaround, optimizer/filter changes and schedule choices are separately declared in Sessions F–H.

Resident libCEED experiment

The tested library is v0.12.0, commit 4018a20a98d451fac24765d3ddb936861647ce8d, archive SHA-256 4c642ae1563e0363aa48f5d794832ad6e9716789405e33511213b837d1528a23. evidence/companion/results/rescope/hex-modal-libceed-resident-20260924/protocol.json pins the source and configuration; no current-library or unreleased-main claim is made.

The extracted bridge and QFunction are retained as provenance/supplement/libceed-bridge.c and libceed-elasticity.h, with archive/member hashes in provenance/supplement/extractions.json. The bridge creates a 3D, three-component Q1 basis with two Gauss points per axis. A discontinuous constant scalar basis stores one density per cell; the QFunction evaluates $E=1e-6+(1-1e-6)\rho^3$, with unit modulus, Poisson ratio 0.3 and exact unit-cube geometry scaling. It does not store eight replicated density coefficients per cell. Library nodes are (z,y,x,component) while the custom padded layout is (x,z,y,component); a reflected-y restriction maps the same physical nodes. Constraints are exact identity rows imposed by a GPU kernel after the library action.

The requested backends are /gpu/cuda/ref, /gpu/cuda/shared and /gpu/cuda/gen, with device-memory preference checked. One action includes output clearing/assembly, the density-scaled elastic law, identity rows and cudaDeviceSynchronize. Host steady-clock batched time is primary; CUDA-event time is secondary. Setup, conversion, allocation, transfers, compilation, JIT warmup and numerical output are excluded. Input/material vectors remain resident. There are 12 randomized blocks of 256 calls, five seconds of warmup, and a minimum five-millisecond batch criterion. Six primary cases are 64×32×32, 128×64×64 and 192×96×96, each with two coefficient patterns; 18 total shapes/patterns and separate sanitizer runs receive correctness checks.

The 2.5768–3.0083 ratio is generated-backend time divided by signed-multiply-add time for those six cases (figures/ratios.csv; regenerate via analysis/session_e_tables.py). It comes from one RTX 4090 allocation and a different harness from the main CuPy study. It does not establish superiority over libCEED as a whole, other backend choices, or tensor implementations in general. The declared sign-test/Holm rule covers all 30 case/control comparisons and must not be replaced with Session E's paired-bootstrap rule.

Pinned release audit

The detailed finding table is [the reproducibility record](#). companion/NAME there abbreviates evidence/companion/results/rescope/NAME. The OpenMP pin is b46ed453ad5a05818b563e540ce176c9d85c6fe2; the Futhark pin is c3cd2b00632ad57cd3d8917246552b2fb4e99e01. Their repository URLs, commits and per-file hashes are retained in provenance/supplement/traff-source-provenance.json. These are pinned GitHub trees; the original source-provenance record does not establish byte identity with the authors' separate Zenodo version archive. The source pins and any adapted-source hashes are in each row's frozen protocol.

The 12.17% value is an initial candidate integration error. The archived diagnosis.json identifies published stencil_methods.c:426 as copying input on fixed rows, while the first candidate's interpose.c set those rows to zero. Free-DOF probes could not distinguish them. The initial CPU experiment has 18 rejected state gates, including small nonzero fixed displacements from the source and much larger candidate errors. Its result must not be presented as a 12.17% error in the unmodified published solver, or as the result of a candidate that already honored identity rows. The next CPU experiment restored candidate identity rows and added shared exact projection of returned fixed values; its 18/18 passing states concern that declared adaptation.

The diagnosis has been copied byte-for-byte to provenance/supplement/host-diagnosis.json from the deposited companion--hex-modal-mg-host-20260923--files.tar.gz, whose SHA-256 is recorded in the extraction manifest. No original evidence file has been overwritten.

Quantity	Definition, denominator and units	Input and interpretation
12.170770009% state error	$100 \cdot \text{norm}(u - u_{\text{ref}}, 2) / \text{norm}(u_{\text{ref}}, 2)$, full DOFs; dimensionless percentage; direct constrained source-matched reference	Initial modal candidate, $24 \times 12 \times 8$, step 1. Rescaling a linear right-hand side scales both numerator and denominator; conditioning and rounding can still change observed relative error.
1.139422914e-5 bilinear defect	$\text{abs}(a^{\text{TB}}(b) - b^{\text{TB}}(a)) / (\text{norm}(a) \cdot \text{norm}(B(b)) + \text{norm}(b) \cdot \text{norm}(B(a)))$; dimensionless	CPU source-transfer V-cycle, $16 \times 8 \times 8$, uniform density, pair 2. A sampled reciprocity error, sensitive to finite-precision summation and cancellation; not a displacement error or a universal SPD proof.
66.593264989 fixed-DOF leakage	$\max(\text{abs}(B(a)[\text{fixed}]), \text{abs}(B(b)[\text{fixed}]))$; no denominator; absolute correction/displacement units of this benchmark	$24 \times 12 \times 8$, uniform density, pair 0, zero fixed entries in both input residuals. Magnitude scales with input amplitude and material/force units. Nonzero output contradicts the exact free-space-preservation contract; it cannot be compared numerically with the two relative quantities.

python3 analysis/supplement_registry.py verifies the extracted member hashes and recomputes all three reported maxima from their saved vectors, producing audit-units.json. The selected residual probes are bounded by about 0.5 in magnitude; no normalization to unit residual norm was applied. The source-transfer defect and leakage are distinct diagnostics. Projection after restriction/prolongation gives 24/24 passing sampled tests and a maximum defect about $1.19\text{e-}17$; it does not certify arbitrary coefficient fields or alone establish the cause of an earlier GPU failure.

The auxiliary E3 OC diagnosis is confined to experiments/paper4/run_experiments_e1_e10.py:_oc_update at the Galerkin pin: its arithmetic midpoint bisection starts at $[1\text{e-}40, 1\text{e}40]$ and takes 100 steps. Even repeated upper-bound halving leaves $1\text{e}40/2^{100} \approx 7.888609\text{e}9$, preventing the needed small multiplier. The original preprint already identifies E3 divergence and excludes it from its main acceleration evidence. The new contribution is this source-level localization and full-size reproduction, not discovery that the authors' entire optimizer fails. The separate full run_simp OC update is unaffected. Session G uses the explicitly declared filtered top88 update.

Hypothesis convention and chronology

The [generated registry](#) and [full JSON](#) enumerate endpoints, primary cases, thresholds, declaration timestamps, byte-linked execution records and outcomes for Sessions A–J. H3 was not confirmed; H5 was not supported; H11, H11b and H15 failed their recorded decisions. H11b changes the optimizer and step count and remains a separate hypothesis. Packaging and integration gates are separate; H12/H12b are descriptive expectations with work-match qualifications. Therefore “three failed hypotheses” is not a complete whole-study count.

Protocol hashes establish identity. The registry checks the recorded declaration time against the start of the job that used the same protocol bytes (or a verifier’s matching protocol hash). These are retained author/system clock records, not independent timestamp certification or proof against retrospective authorship. Session J’s final declaration must only be paired with attempt 07; earlier attempts used their separately preserved amendments. Historical release tags identify artifact versions, not proof of prospective declaration. This chronology assessment uses the retained records rather than a reconstruction of Git history.

Principal-claim evidence map

The initial software release is [Zenodo 23001863](#); raw evidence is [Zenodo 22998404](#). Tags below are historical release locators listed in the artifact README. The inspected source baseline is a60700fe9df55d4b2e597e4a4014cee189ac5e43. This supplement records subsequent evidence checks and regenerated summaries; the historical tags identify the original experiment snapshots.

Claim / retained failure	Historical tag	Protocol and raw/summarized evidence	Exact analysis command (repository root)
Exact parity operator and local lane equivalence	exp01-parity-algebra, base	code/hex_modal/protocol.json; code/hex_modal/modal.py; CPU tests; algebra supplement	python3 -m pytest -q tests
Plain FP64 product on RTX 4090 and independent RTX 3090	exp02-products-rtx4090, exp03-products-rtx3090	code/hex_modal_cuda/protocol.json; evidence/kernel-rtx4090/, evidence/kernel-rtx3090-replication/; paper/numbers.json	python3 analysis/paper_numbers.py
Fixed-work PCG and no useful FP32 parity gain; failed A100 dense-control prediction	exp07-session-a, exp08-session-b	experiments/session_{a,b}/protocol.json; results/session-{a,b}-20260923/local-verification.json; paper/session_{a,b}_numbers.json	python3 analysis/session_a_tables.py results/session-a-20260923; python3 analysis/session_a_tables.py results/session-b-20260923 session_b
Tested high-contrast FP64-only niche unsupported (H5)	exp09-session-c	experiments/session_c/protocol.json; results/session-c-20260923/local-verification.json	python3 analysis/session_a_tables.py results/session-c-20260923 session_c
Rediscretized multigrid route comparison	exp10-session-d	experiments/session_d/protocol.json; results/session-d-20260923/local-verification.json	python3 analysis/session_d_tables.py results/session-d-20260923
Two-host FP64 product ratio 2.17–2.25 and solver effects; narrow libCEED result	exp11-session-e	experiments/session_e/protocol.json; results/session-e-host{1,2}-20260925/local-verification.json; companion libCEED protocol/ratios above	python3 analysis/session_e_tables.py

Claim / retained failure	Historical tag	Protocol and raw/summarized evidence	Exact analysis command (repository root)
Packaging failure, integration gate failure, Galerkin capacity	exp12-session-f	experiments/session_f/protocol.json; results/session-f-*/remote/{job.json, run-01/result.json}; paper/session_f_summary.json	python3 analysis/session_f_summary.py
30-step and capped 150-step trajectories; failed H11	exp13-session-g	experiments/session_g/protocol.json; results/session-g-host{1,2}-20260926/remote/run-01/result.json	python3 analysis/session_g_tables.py results/session-g-host1-20260926 results/session-g-host2-20260926
All four 100-step cases and capped 150-step outcomes; failed H11b	exp14-session-h	experiments/session_h/protocol.json; results/session-h-host{1,2}-20260926/remote/run-01/result.json	python3 analysis/session_h_tables.py results/session-h-host1-20260926 results/session-h-host2-20260926
Measured capacity and kernel-map exceptions; RTX fixed-work FP64 gain ~1.35 at 43 M DOFs	exp15-session-i	experiments/session_i/protocol.json; results/session-i-*/remote/run-01/result.json; paper/session_i_numbers.json	python3 analysis/session_i_tables.py results/session-i-rtx4090-20260926 results/session-i-a100-20260926 results/session-i-h100-20260926
Executed counter ratio 2.5; H15 count failure; post-hoc timing agreement	exp16-session-j	experiments/session_j/protocol.json; attempt-07 remote/ncu/ncu-*.csv; all failed attempts retained	python3 analysis/session_j_counters.py
Pinned quadrature, boundary, transfer and damping audit	exp17-traff-audit	evidence/companion/results/escape/hex-modal-*/protocol*.json and result/replay files in reproducibility record; selected deposited inputs in prove nance/supplement/	python3 analysis/repro_numbers.py; python3 analysis/supplement_registry.py

Brace and wildcard paths above denote families, not literal filenames. The registry generator reads the freshly regenerated summaries; run the combined numerical build before it when other analysis sources have changed.

Reproducibility levels and bibliography metadata

Numerical regeneration reads retained results and recomputes summaries, tables, macros and figures on a CPU. **Saved-state replay** reloads recorded coefficient/displacement arrays and checks a reference operator or acceptance decision; it is not a fresh performance measurement. **Fresh execution** recreates the declared GPU software/hardware environment, runs the frozen workload and collects new timings. The public repository supports numerical regeneration; larger saved-state and raw-source members are accessible in the existing deposits through their manifests. The manuscript text and its build are not part of the public repository.

The [PLOS publisher record](#) confirms volume **15(10)** and DOI 10.1371/journal.pone.0240813. The [Springer publisher record](#) gives **Long Hu, Tianyuan Qi, Junpeng Zhao and Chunjie Wang**, volume 69, article 90 and DOI 10.1007/s00158-026-04280-3. These fields are recorded in paper/manuscript/refs.bib; the records were checked on 2026-09-27. The linked September 2026 Hu correction concerns interchanged ellipse semi-axes in Fig. 25 and does not change these

bibliography fields.

Hypothesis and gate registry

Five named hypotheses lack support: H3, H5, H11, H11b, H15. H5 retains its original not-supported label; H11b is a separate changed-pipeline hypothesis. Descriptive H12/H12b expectations and packaging/integration gates are separate. This count is for Sessions A-J, not every exploratory companion study.

Hashes identify content, not temporal priority. These dates are author/system recorded timestamps, checked against byte-linked execution records where available, not independent preregistration timestamps. No local git history is present in this workspace.

Frozen Session B h3.rule writes modal8/dense8, but the question and recorded analysis concern advantage dense8 time / modal8 time. The retained failed decision uses that latter ratio (about 2.24), not the literal reversed label. The historical protocol is preserved.

The JSON companion preserves the exact primary cases, decision rules, hashes and every inspected execution record. The table abbreviates these fields.

ID	Type	Cases / endpoint and decision	Declaration UTC	Executed UTC (byte-linked final protocol)	Outcome
H2	hypothesis	optimized q64/q96; fixed Jacobi-PCG work, median ≥ 1.08 and one-sided 95% lower ≥ 1.05	2026-09-23 17:15:38	2026-09-23 17:26:40	supported
H3	hypothesis	optimized q64/q96; FP64 dense8/parity product gain < 1.10 expected	2026-09-23 18:05:36	2026-09-23 18:41:35	not confirmed
H4	hypothesis	optimized q64/q96; FP32 gain over fastest baseline < 1.10 expected	2026-09-23 17:15:38	2026-09-23 17:26:40	supported
H5	hypothesis	optimized q64/q96 at $E_{min}=1e-9$; at least one refinement failure while all FP64 PCG repetitions pass	2026-09-23 18:38:14	2026-09-23 18:39:47	not supported
H6	hypothesis	optimized q64/q96; fixed MG work, median ≥ 1.05 , lower ≥ 1.02	2026-09-23 19:35:12	2026-09-23 19:38:27	supported
H7	descriptive question	fastest physically accepted solve to tolerance; descriptive	2026-09-23 19:35:12	2026-09-23 19:38:27	descriptive
H8	hypothesis	optimized q64/q96 on both hosts; product median ≥ 1.05 , lower ≥ 1.02	2026-09-25 10:54:43	2026-09-25 10:58:45, 2026-09-25 11:09:30	supported on two hosts
H9	hypothesis	same cases; fixed MG work median ≥ 1.05 , lower ≥ 1.02	2026-09-25 10:54:43	2026-09-25 10:58:45, 2026-09-25 11:09:30	supported on two hosts

ID	Type	Cases / endpoint and decision	Declaration UTC	Executed UTC (byte-linked final protocol)	Outcome
H10	descriptive question	same cases; fastest solve and mixed outer-product effect, descriptive	2026-09-25 10:54:43	2026-09-25 10:58:45, 2026-09-25 11:09:30	descriptive
F-stack	packaging gate	three GPUs; unchanged donor smoke-test exit code 0	2026-09-26 00:42:40	2026-09-26 00:59:35, 2026-09-26 01:18:10, 2026-09-26 00:44:22	failed as released; directory workaround passes
F-integration	integration gate	all grids, densities, schedules and arms; product $\leq 1e-12$, convergence, residual $\leq 1e-10$, cross-state $\leq 1e-8$	2026-09-26 00:42:40	2026-09-26 00:59:35, 2026-09-26 01:18:10, 2026-09-26 00:44:22	not passed
H11	hypothesis	three primary cases, FP64 30-step trajectories; all medians ≥ 1.10 , ≥ 2 repetitions, work/agreement gates	2026-09-26 01:41:32	2026-09-26 01:44:43, 2026-09-26 02:44:46	failed on two hosts
H12	descriptive expectation	same primary cases, mixed trajectories; descriptive expectation < 1.10	2026-09-26 01:41:32	2026-09-26 01:44:43, 2026-09-26 02:44:46	not uniformly below 1.10; work-match qualification required
H11b	new hypothesis on changed pipeline	same three primary cases, GPU optimizer, 100 steps; same ≥ 1.10 and qualification rules	2026-09-26 04:15:46	2026-09-26 04:17:38, 2026-09-26 04:18:54	failed on two hosts
H12b	descriptive expectation	GPU optimizer mixed trajectories; descriptive expectation < 1.10	2026-09-26 04:15:46	2026-09-26 04:17:38, 2026-09-26 04:18:54	not uniformly below 1.10; work-match qualification required
H13	hypothesis	fitting primary enlarged cases on each 80 GB card; parity/node fixed-work time $\geq 1/1.05$	2026-09-26 04:22:52	2026-09-26 04:40:45, 2026-09-26 05:27:10, 2026-09-26 04:30:46	supported on fitting primary cases
H14	hypothesis	largest fitting primary RTX case; fused/parity fixed-work median ≥ 1.18	2026-09-26 04:22:52	2026-09-26 04:40:45, 2026-09-26 05:27:10, 2026-09-26 04:30:46	supported at largest fitting primary case
H15	hypothesis	128×64×64 RTX4090 VM; FP64 active utilization $\geq 70\%$; executed count ratios within 5% of static 78:52:34	2026-09-26 05:25:58	2026-09-26 06:04:06	failed count criterion; utilization part passed

Reproducibility record for the combined paper

Consolidates the audit findings on the published codes used as baselines and hosts. Each row gives the finding, how it was established, what repaired it (if anything), and the sealed evidence. Every evidence entry is a path in this repository: the Träff et al. audit files under `evidence/companion/` (abbreviated `companion/` below means `evidence/companion/results/rescope/`; their protocols and results keep their original layout) and the session folders under `results/`. Declared protocols and dated execution records are linked in [the hypothesis registry](#). Hashes establish content identity; the recorded timestamps support chronology without constituting independent preregistration. Failures are kept.

These findings are about specific pinned releases and paths. They are not claims about other versions, and they do not say whether the published papers' conclusions hold.

A. Träff et al. topology-optimization codes (OpenMP GPU and Futhark releases)

#	Finding	Established by	Repair / consequence	Evidence
A1	The OpenMP release uses composite 10×10×10 midpoint quadrature per fine element : <code>getKESubspace</code> samples ten points per fine-subcell axis. Relative Frobenius differences from exact 2×2×2 Gauss Q1 are 0.548361% / 0.516454% / 0.416469% at $\nu = 0.20 / 0.30 / 0.45$. The tested composite matrices retain six rigid modes; this is not the eighteen-mode pathology of literal single-point Hex8. The pinned Futhark matrix matches exact Q1.	Independent exact-Q1/source-matched composite integration replay on pinned sources (OpenMP b46ed453, Futhark c3cd2b00); fresh CPU quadrature and eigenvalue checks in <code>paper/supplement/algebra-checks.json</code> use null tolerance $1e-10$ <code> K _2</code> .	This is a mismatch with this study's exact-Gauss operator contract, not evidence by itself that the audited mechanics or original conclusions are invalid. Baselines in this study use exact Gauss Q1; composite-rule comparisons are labeled source-matched.	<code>companion/hex-modal-priorart-20260923/</code> (<code>cpu-protocol-02.json</code> , <code>cpu-02/results.json</code>); quadrature definitions and check
A2	Both releases run mixed precision (FP32 vectors, correction or CG in parts), not all-FP64. The pinned trees contain no licence file.	Source trace	All-FP64 claims use separately declared adaptations. No redistribution permission is inferred.	pinned sources (commits above)

#	Finding	Established by	Repair / consequence	Evidence
A3	Initial candidate boundary mismatch: the candidate zeroed fixed rows instead of copying the input as the published operator does; its state error reached 12.17 % . Separately, source runs leave constrained displacement up to 5.27e-7 , violating this audit's exact-zero acceptance gate. Free-input probes cannot distinguish zero from identity fixed rows.	Serial CPU execution of source C with OpenMP pragmas ignored; 18 captured solves on 16×8×8, 24×12×8 and 16×8×12 over three steps; all 36 source/candidate free-product probes pass, all 18 state gates fail	Candidate identity rows are restored, and a common returned-state projection is declared for both arms. The adapted CPU experiment passes 18/18 states, 9/9 paired and 3/3 endpoint comparisons. 12.17 % is not an unmodified-source solver error.	companion/hex-modal-mg-host-20260923/{protocol.json,run-01/result.json}; archived diagnosis now at provenance/supplement/host-diagnosis.json; companion/hex-modal-mg-host-v2-20260923/
A4	Source transfer boundary contract: unprojected restriction/prolongation in the declared adapted host fail 24/24 sampled free-space/symmetry probes (dimensionless bilinear defect up to 1.14e-5; absolute fixed correction leakage 66.6).	CPU execution of C VcyclePreconditioner, not an original GPU rerun; three meshes, uniform/random density, four residual pairs each	Zero fixed outputs after each restriction and prolongation: 24/24 pass, dimensionless defect $\leq 1.19\text{e-}17$. This tests necessary properties, not universal SPD or causation of an earlier GPU failure.	companion/hex-modal-mg-bc-audit-20260923/{protocol.json,run-01/verification.json}; exact normalization and saved-array replay in paper/supplement/provenance-audit.md
A5	Damping: fixed weighted-Jacobi $\omega=0.6$ admits unstable modes at level two on retained designs under this study's exact-Q1 operator contract.	CPU independent spectral diagnostic: 9/9 converged eigen-estimates; exact local bounds on 73 subcell matrices; subsequently a separately declared GPU replay	Weights [0.3,0.25,0.2] have exact rational LDL^T local certification for ideal Q1. The matched GPU replay accepts 9/9 saved states versus 3/9 with $\omega=0.6$. This is not a universal mixed-precision convergence proof.	companion/hex-modal-mg-smooth-audit-20260923/; companion/hex-modal-mg-damping-control-20260923/run-03/exact-summary.json; companion/hex-modal-mg-damping-gpu-20260923/rental-01/local-verification.json

B. Yang et al. Galerkin GMG release (nbbllxx0/Mixed-Precision-GMG-SIMP, fd9ddec)

#	Finding	Established by	Repair / consequence	Evidence
B1	Package/path-discovery failure: paths.py requires a scripts/ directory beside src/, absent from the pinned public tree.	Original ci/smoke_test.py attempted on RTX 4090, A100 PCIe 80 GB and H100 NVL; nonzero exit before numerical smoke solve	Creating an empty directory allows smoke tests to run and pass. This is a packaging workaround, not repair of a numerical algorithm. RTX smoke: 25 iterations, residual 5.95e-11.	experiments/combined/donor_yang_gmg/src/gpu_fem/paths.py; results/session-f-*/remote/job.json; paper/session_f_summary.json

#	Finding	Established by	Repair / consequence	Evidence
B2	The auxiliary E3 loop's OC update collapses the design (confirmed at full size on two RTX 4090 hosts, session G: volume 0.30 → 0.15 → 0.001, compliance → 6.9e8). E3's own <code>_oc_update</code> bisects the multiplier arithmetically over [1e-40, 1e40] for 100 iterations, so it cannot fall below ≈ 7.9e9. Volume went 0.30 → 0.15 → 0.001 and compliance 140 → 1.9e9 on a CPU replica of their cantilever. The paper already treats E3 as auxiliary: "not used as main acceleration evidence because the design trajectories diverge" (§4.4, App. A). It attributes the divergence to capped Jacobi solves, and its E3 compliance plots are at 10^8–10^9. Their full SIMP driver (<code>run_simp</code>) uses a correct OC (bisection on [0, 1e9] to 1e-9, volume on filtered/projected densities) and is not affected.	CPU replica; paper full text read; full-size diagnostic in session G	Only the auxiliary E3 schedule is affected; the paper's headline claims do not rest on it. Session G uses a top88 OC with a density filter.	session G (<code>results/session-g-*</code>); arXiv 2604.26441 §4.4, App. A
B3	FP64 comparator provenance: the Galerkin stock core is CuPy gather + DGEMM + <code>bincount</code>; its fused kernels are FP32/BF16.	Pinned <code>solver_v2.py:MatrixFreeKff.matvec</code> and <code>cuda_fused_matvec.py</code> source	The supplied FP64 fused and node paths come from the separate fused-kernel release at 3a6e37af137c3f8ba65ef1d298602a6390da6466. The FP64 core substitution is an explicit adapter; it does not make the complete solver unmodified.	<code>experiments/combined/kff_adapters.py</code> ; <code>experiments/session_f/protocol.json</code> ; comparator symbol/hash table in <code>paper/supplement/provenance-audit.md</code>
B4	Measured allocation footprint: about 23 GiB per million elements for this pinned Galerkin implementation. 2 M fits, whereas 5 M fails on A100 PCIe 80 GB and H100 NVL 94 GB.	Session F sizing ladder, per schedule/arm, including out-of-memory outcomes	About 3 M is an extrapolated ceiling, not a tested maximum. These records do not fully separate live arrays, transient construction allocations and cached pool memory. Larger studies use the separately specified rediscretized hierarchy.	<code>experiments/session_f/protocol.json</code> ; <code>results/session-f-*/remote/run-01/result.json</code> ; <code>paper/session_f_summary.json</code>

#	Finding	Established by	Repair / consequence	Evidence
B5	Mixed-schedule integration gate: the stock FP32 fine-smoother schedule reaches 300 iterations on random-density 64 k cantilever states on A100/H100; stock takes 246 on RTX 4090. Other adapted arms also fail parts of the compound gate.	Original and adapted GPU solver execution in Session F; product agreement, convergence, residual and cross-state requirements recorded separately	The gate remains failed on every host despite FP64 core product agreement $\sim 2.2e-16$. The donor paper already discloses difficult heterogeneous convergence; this audit localizes failure in its own declared workload, not a new universal defect. Session G changes to SIMP-generated trajectories and records non-converged solves.	experiments/session_f/protocol.json; results/session-f-*/remote/run-01/result.json; paper/session_f_summary.json

C. Process and units

The [supplement](#) defines the state-error denominator, normalized bilinear defect and absolute correction leakage, and reproduces their selected maxima from hash-verified saved arrays. The three quantities have different units and cannot be compared as percentages. In particular, the magnitude of absolute fixed-DOF leakage scales with the residual input; preservation of the free space is the violated invariant.

The [registry](#) distinguishes named hypotheses from descriptive expectations, packaging failures, integration gates and operational attempts. H3 was not confirmed, H5 was not supported, and H11/H11b/H15 failed their recorded decisions. H11b concerns a changed pipeline; it does not erase H11. No global count of three failures is claimed.

GPU operational failures (upload timeouts, rate limits, SSH, startup, counter permission and tooling) remain separate attempts, including Session J's seven attempts. Each final protocol is linked to matching execution records; earlier failed attempts use their own amendments. Hashes alone do not prove temporal priority. Historical thresholds and outcomes remain preserved. New adaptations are explicitly labeled, including the boundary projection, corrected OC and GPU optimizer.

D. Source, input and contract locator

This table supplements the observed-error/adaptation/outcome columns above. Source paths are relative to each pinned upstream tree unless identified as an audit script. The Träff host/transfer entries are CPU source execution; the damping follow-up and Yang sessions explicitly executed GPUs.

Finding	Pinned source / location	Exercised input and mathematical contract	Evidence type / result locator
A1 quadrature	OpenMP b46ed453..., local_matrix.c :getKESubspace; Futhark c3cd2b00..., src/keConstants.fut	Three Poisson ratios, levels/subcells in cpu-protocol-02.json; exact-Gauss Q1 comparison, not whole-element reduced integration	CPU matrix comparison; hex-modal-priorart-20260923/cpu-02/results.json
A2 precision	Same pins; OpenMP definitions.h, solver vectors; Futhark src/solvers.fut	Source scalar/vector types versus this study's FP64-throughout contract	Source trace, not a speed claim; source hashes in A1 protocol

Finding	Pinned source / location	Exercised input and mathematical contract	Evidence type / result locator
A3 boundary	OpenMP b46ed453..., stencil_method s.c:426; initial audit interpose.c	Three meshes × three steps × source/candidate; source identity rows versus candidate zero rows; full-state norm and exact fixed values	CPU source execution; diagnosis and raw/saved state checks above
A4 transfer	Same base, declared adapted-02 source hashes; stencil_solvers.c:Vcycle Preconditioner, restriction/prolongation wrappers	16×8×8, 24×12×8 (3 levels), 32×16×16 (4); uniform 0.2/random void patterns; free-space preservation and reciprocity	CPU C source transfers; hex-modal-mg- bc-audit-20260923 protocol and verification
A5 damping	Same base; fixed-Jacobi source semantics, independent spectrum.py audit	Retained 128×64×64 states 8/9 and 64×32×32 state 21; Jacobi stability and subsequently true-residual/state gates	CPU spectral and rational certification, then separately adapted GPU replay; three audit folders above
B1 path discovery	Galerkin fd9ddecd..., src/gpu_fem/paths.py, ci/smoke_test.py	Three GPU hosts; package root discovery and zero-exit smoke gate	Original GPU-host launch failure, then directory-only workaround
B2 auxiliary OC	Same pin, experiments/paper4/run_ experiments_e1_e10.py:_oc_upda te	Auxiliary E3 216k cantilever, mixed stock schedule; multiplier bisection must span volume-preserving update	CPU arithmetic diagnosis; Session G full-size GPU diagnostic on two hosts; full run_simp unaffected
B3 FP64 core	Galerkin fd9ddecd...:solver_v2.py; fused 3a6e37af...:cuda_operators.p y	Same caller-supplied Ke, modulus and constraints, stock versus substituted fine actions	Supplied-code identity plus explicit adapter and GPU product checks
B4 memory	Galerkin fd9ddecd..., multigrid_v4.py	2 M/5 M element ladder; allocation feasibility by schedule/arm	GPU sizing, not exact capacity limit or universal Galerkin-memory bound
B5 integration	Same pin, solver_v4.py / mixed fine smoother	Small/medium uniform and seeded random fields; product ≤1e-12, residual ≤1e-10, convergence and cross-state ≤1e-8	Original/adapted GPU runs; failed compound gate retained

Optimization outcomes and scale tables

The tables regenerate from retained raw records. NR means not recorded, not zero. Exact per-repetition records, source paths and JSON pointers are in optimization-outcomes.csv, scale-solves.csv and outcomes-scale.json. No new GPU experiment is represented here. Host keys G1/G2 and H1/H2 identify the two RTX 4090 hosts in Sessions G/H. IA/IH/IR identify Session I A100 PCIe, H100 80GB HBM3 and RTX 4090; FA/FH/FR identify Session F A100 PCIe, H100 NVL and RTX 4090. C216/C512 are cantilevers, M514 the MBB case and T499 torsion. S = stock gather/DGEMM/scatter, F = fused-ai FP64, P = parity mul-add, N = node FP64. Schedule 64 = all FP64; 32 = mixed. Scale arms 64F/64N/64P use FP64 throughout; 32F/32N/32P use a node FP32 V-cycle with the named FP64 outer product.

Every optimization case, schedule and arm

Each fixed-length row aggregates three repetitions by separate column medians. Capped rows and auxiliary diagnostics have n=1. Exit L = prescribed fixed length; C = 150-step iteration cap; D = auxiliary OC diagnostic. K is total Krylov iterations. An L exit is not a convergence test. Every C exit missed design change 0.01. O IDs match across the two tables.

Table 1: Optimization identity, stopping outcome and wall time. Failed-solve count is summed over repetitions.

ID	Host	Case	Sch.	Arm	n	Steps	Exit	K	Wall s	Failed
O1	G1	C216	64	S	3	30	L	790	20.263	0
O2	G1	C216	64	F	3	30	L	790	16.742	0
O3	G1	C216	64	P	3	30	L	790	16.561	0
O4	G1	C216	32	S	3	30	L	842	18.259	0
O5	G1	C216	32	P	3	30	L	847	17.063	0
O6	G1	C216	32	F	3	30	L	846	17.029	0
O7	G1	C512	64	P	3	30	L	404	34.927	0
O8	G1	C512	64	F	3	30	L	401	35.794	0
O9	G1	C512	64	S	3	30	L	401	40.001	0
O10	G1	C512	32	P	3	30	L	411	34.882	0
O11	G1	C512	32	F	3	30	L	413	35.119	0
O12	G1	C512	32	S	3	30	L	419	36.693	0
O13	G1	M514	64	S	3	30	L	1829	63.588	0
O14	G1	M514	64	F	3	30	L	1848	45.832	0
O15	G1	M514	64	P	3	30	L	1836	42.925	0
O16	G1	M514	32	F	3	30	L	3510	63.561	2
O17	G1	M514	32	P	3	30	L	3378	54.234	0
O18	G1	M514	32	S	3	30	L	3589	64.722	2
O19	G1	T499	64	F	3	30	L	1047	37.838	0
O20	G1	T499	64	S	3	30	L	1038	47.701	0
O21	G1	T499	64	P	3	30	L	1020	36.198	0
O22	G1	T499	32	P	3	30	L	1170	36.555	0
O23	G1	T499	32	S	3	30	L	1157	40.798	0
O24	G1	T499	32	F	3	30	L	1166	37.093	0
O25	G1	C216	64	S	1	150	C	3416	80.809	0
O26	G1	C216	64	F	1	150	C	3376	65.638	0
O27	G1	C216	64	P	1	150	C	3397	65.169	0
O28	G1	C216	32	S	1	150	C	3467	71.677	0
O29	G1	C216	32	F	1	150	C	3539	66.762	0
O30	G1	C216	32	P	1	150	C	3478	65.924	0
O31	G1	C216	32	S	1	30	D	339	11.811	0
O32	G2	C216	64	S	3	30	L	790	21.939	0
O33	G2	C216	64	F	3	30	L	790	17.981	0
O34	G2	C216	64	P	3	30	L	790	18.037	0
O35	G2	C216	32	S	3	30	L	836	19.534	0
O36	G2	C216	32	P	3	30	L	844	18.390	0
O37	G2	C216	32	F	3	30	L	840	18.028	0
O38	G2	C512	64	P	3	30	L	404	37.917	0
O39	G2	C512	64	F	3	30	L	403	39.110	0
O40	G2	C512	64	S	3	30	L	404	43.051	0
O41	G2	C512	32	P	3	30	L	413	37.768	0

ID	Host	Case	Sch.	Arm	n	Steps	Exit	K	Wall s	Failed
O42	G2	C512	32	F	3	30	L	415	38.151	0
O43	G2	C512	32	S	3	30	L	410	40.037	0
O44	G2	M514	64	S	3	30	L	1843	67.523	0
O45	G2	M514	64	F	3	30	L	1827	48.714	0
O46	G2	M514	64	P	3	30	L	1820	45.753	0
O47	G2	M514	32	F	3	30	L	3801	74.386	2
O48	G2	M514	32	P	3	30	L	2905	56.535	0
O49	G2	M514	32	S	3	30	L	2949	71.794	0
O50	G2	T499	64	F	3	30	L	1034	41.367	0
O51	G2	T499	64	S	3	30	L	1029	51.287	0
O52	G2	T499	64	P	3	30	L	1025	39.579	0
O53	G2	T499	32	P	3	30	L	1153	40.045	0
O54	G2	T499	32	S	3	30	L	1162	43.891	0
O55	G2	T499	32	F	3	30	L	1171	40.225	0
O56	G2	C216	64	S	1	150	C	3395	84.020	0
O57	G2	C216	64	F	1	150	C	3396	69.404	0
O58	G2	C216	64	P	1	150	C	3402	68.641	0
O59	G2	C216	32	S	1	150	C	3585	75.701	0
O60	G2	C216	32	F	1	150	C	3495	71.565	0
O61	G2	C216	32	P	1	150	C	3463	74.958	0
O62	G2	C216	32	S	1	30	D	341	12.437	0
O63	H1	C216	64	S	3	100	L	2257	27.661	0
O64	H1	C216	64	F	3	100	L	2252	17.540	0
O65	H1	C216	64	P	3	100	L	2244	17.048	0
O66	H1	C216	32	S	3	100	L	2316	21.346	0
O67	H1	C216	32	P	3	100	L	2323	17.674	0
O68	H1	C216	32	F	3	100	L	2354	17.932	0
O69	H1	C512	64	P	3	100	L	1244	26.272	0
O70	H1	C512	64	F	3	100	L	1253	28.593	0
O71	H1	C512	64	S	3	100	L	1251	41.354	0
O72	H1	C512	32	P	3	100	L	1263	26.062	0
O73	H1	C512	32	F	3	100	L	1260	26.665	0
O74	H1	C512	32	S	3	100	L	1262	31.278	0
O75	H1	M514	64	S	3	100	L	5298	109.817	0
O76	H1	M514	64	F	3	100	L	5305	58.233	0
O77	H1	M514	64	P	3	100	L	5246	49.298	0
O78	H1	M514	32	F	3	100	L	8469	79.077	2
O79	H1	M514	32	P	3	100	L	9044	74.400	2
O80	H1	M514	32	S	3	100	L	8194	119.997	3
O81	H1	T499	64	F	3	100	L	1890	29.155	0
O82	H1	T499	64	S	3	100	L	1890	47.322	0
O83	H1	T499	64	P	3	100	L	1889	25.766	0
O84	H1	T499	32	P	3	100	L	2064	26.172	0

ID	Host	Case	Sch.	Arm	n	Steps	Exit	K	Wall s	Failed
O85	H1	T499	32	S	3	100	L	2078	34.564	0
O86	H1	T499	32	F	3	100	L	2070	27.353	0
O87	H1	C216	64	S	1	150	C	3398	39.744	0
O88	H1	C216	64	F	1	150	C	3387	24.184	0
O89	H1	C216	64	P	1	150	C	3388	23.689	0
O90	H1	C216	32	S	1	150	C	3502	29.985	0
O91	H1	C216	32	F	1	150	C	3443	24.592	0
O92	H1	C216	32	P	1	150	C	3689	28.915	0
O93	H2	C216	64	S	3	100	L	2247	27.828	0
O94	H2	C216	64	F	3	100	L	2251	17.762	0
O95	H2	C216	64	P	3	100	L	2258	17.345	0
O96	H2	C216	32	S	3	100	L	2355	21.816	0
O97	H2	C216	32	P	3	100	L	2319	18.153	0
O98	H2	C216	32	F	3	100	L	2303	18.438	0
O99	H2	C512	64	P	3	100	L	1249	26.217	0
O100	H2	C512	64	F	3	100	L	1252	28.376	0
O101	H2	C512	64	S	3	100	L	1251	41.178	0
O102	H2	C512	32	P	3	100	L	1252	25.912	0
O103	H2	C512	32	F	3	100	L	1258	26.652	0
O104	H2	C512	32	S	3	100	L	1257	31.149	0
O105	H2	M514	64	S	3	100	L	5335	109.297	0
O106	H2	M514	64	F	3	100	L	5275	57.495	0
O107	H2	M514	64	P	3	100	L	5288	49.275	0
O108	H2	M514	32	F	3	100	L	8746	88.197	2
O109	H2	M514	32	P	3	100	L	8493	73.942	2
O110	H2	M514	32	S	3	100	L	8208	110.312	2
O111	H2	T499	64	F	3	100	L	1891	28.776	0
O112	H2	T499	64	S	3	100	L	1873	47.041	0
O113	H2	T499	64	P	3	100	L	1889	25.805	0
O114	H2	T499	32	P	3	100	L	2051	25.990	0
O115	H2	T499	32	S	3	100	L	2106	34.248	0
O116	H2	T499	32	F	3	100	L	2079	27.208	0
O117	H2	C216	64	S	1	150	C	3373	39.403	0
O118	H2	C216	64	F	1	150	C	3385	24.448	0
O119	H2	C216	64	P	1	150	C	3373	23.930	0
O120	H2	C216	32	S	1	150	C	3747	31.718	0
O121	H2	C216	32	F	1	150	C	3469	25.064	0
O122	H2	C216	32	P	1	150	C	3477	24.815	0

CG and re-setup are sums of per-step timers, then medians across repetitions. Initial hierarchy construction is included in wall time but separately NR for every row. Logged compliance precedes the final OC update; volume and change follow it. Change is the maximum absolute unfiltered design-variable update. Density-field agreement was not computed.

Table 2: Optimization solve/setup and final logged quantities (O IDs as above).

ID	CG s	Re-setup s	Compliance	Volume	Change
O1	6.634	0.852	2.407936	0.299996	0.150000
O2	3.150	0.774	2.407936	0.299996	0.150000
O3	3.021	0.775	2.407936	0.299996	0.150000
O4	4.630	0.853	2.407936	0.299996	0.150000
O5	3.372	0.777	2.407936	0.299996	0.150000
O6	3.444	0.782	2.407936	0.299996	0.150000
O7	2.406	1.986	1.966316	0.300003	0.150000
O8	3.056	2.017	1.966316	0.300003	0.150000
O9	7.097	2.206	1.966316	0.300003	0.150000
O10	2.307	1.986	1.966316	0.300003	0.150000
O11	2.547	2.017	1.966316	0.300003	0.150000
O12	3.999	2.207	1.966316	0.300003	0.150000
O13	30.342	2.842	5.796278	0.500000	0.150000
O14	12.831	2.681	5.796278	0.500000	0.150000
O15	9.844	2.655	5.796278	0.500000	0.150000
O16	30.407	2.684	5.796276	0.500000	0.150000
O17	21.133	2.656	5.796276	0.500000	0.150000
O18	31.401	2.844	5.796282	0.499999	0.150000
O19	6.758	1.508	2.084699	0.250001	0.150000
O20	16.480	1.692	2.084698	0.250001	0.150000
O21	5.008	1.478	2.084697	0.250001	0.150000
O22	5.455	1.477	2.084698	0.250001	0.150000
O23	9.606	1.692	2.084695	0.250001	0.150000
O24	5.959	1.508	2.084699	0.250001	0.150000
O25	28.839	3.218	2.370205	0.300000	0.148441
O26	13.670	3.195	2.370205	0.300000	0.148433
O27	13.068	3.255	2.370205	0.300000	0.148441
O28	19.360	3.233	2.370206	0.300000	0.148455
O29	14.601	3.244	2.370205	0.300000	0.148402
O30	13.938	3.256	2.370207	0.300000	0.148475
O31	2.005	1.202	405316117.939373	0.001000	0.000001
O32	6.894	0.895	2.407936	0.299996	0.150000
O33	3.224	0.813	2.407936	0.299996	0.150000
O34	3.140	0.810	2.407936	0.299996	0.150000
O35	4.743	0.889	2.407936	0.299996	0.150000
O36	3.471	0.810	2.407936	0.299996	0.150000
O37	3.418	0.808	2.407936	0.299996	0.150000
O38	2.499	2.079	1.966316	0.300003	0.150000
O39	3.193	2.114	1.966316	0.300003	0.150000
O40	7.379	2.305	1.966316	0.300003	0.150000
O41	2.392	2.072	1.966316	0.300003	0.150000

ID	CG s	Re-setup s	Compliance	Volume	Change
O42	2.637	2.107	1.966316	0.300003	0.150000
O43	4.072	2.307	1.966316	0.300003	0.150000
O44	31.493	2.955	5.796278	0.500000	0.150000
O45	13.088	2.780	5.796278	0.500000	0.150000
O46	10.066	2.760	5.796278	0.500000	0.150000
O47	38.494	2.786	5.796276	0.500000	0.150000
O48	20.103	2.750	5.796279	0.500000	0.150000
O49	35.397	2.945	5.796277	0.500000	0.150000
O50	6.890	1.576	2.084697	0.250002	0.150000
O51	16.846	1.765	2.084697	0.250002	0.150000
O52	5.221	1.549	2.084698	0.250001	0.150000
O53	5.577	1.547	2.084699	0.250001	0.150000
O54	9.874	1.767	2.084696	0.250002	0.150000
O55	6.128	1.573	2.084697	0.250002	0.150000
O56	29.224	3.324	2.370205	0.300000	0.148439
O57	13.806	3.259	2.370205	0.300000	0.148437
O58	13.298	3.295	2.370205	0.300000	0.148443
O59	20.122	3.344	2.370205	0.300000	0.148415
O60	14.987	3.343	2.370206	0.300000	0.148368
O61	14.164	3.319	2.370205	0.300000	0.148403
O62	2.037	1.259	405316068.966563	0.001000	0.000001
O63	18.892	2.252	2.375136	0.300000	0.150000
O64	8.838	2.172	2.375136	0.300000	0.150000
O65	8.383	2.169	2.375136	0.300000	0.150000
O66	12.575	2.257	2.375137	0.300000	0.150000
O67	9.018	2.160	2.375137	0.300000	0.150000
O68	9.322	2.186	2.375137	0.300000	0.150000
O69	7.516	6.245	1.942433	0.299999	0.150000
O70	9.678	6.283	1.942433	0.299999	0.150000
O71	22.392	6.476	1.942433	0.299999	0.150000
O72	7.159	6.258	1.942433	0.299999	0.150000
O73	7.880	6.272	1.942433	0.299999	0.150000
O74	12.264	6.470	1.942433	0.299999	0.150000
O75	88.539	8.763	5.774663	0.500001	0.150000
O76	37.156	8.592	5.774663	0.500001	0.150000
O77	28.391	8.577	5.774663	0.500001	0.150000
O78	58.140	8.608	5.774660	0.499999	0.150000
O79	53.487	8.578	5.774660	0.499999	0.150000
O80	98.228	8.766	5.774661	0.499999	0.150000
O81	12.606	4.577	2.083160	0.250000	0.150000
O82	30.818	4.758	2.083158	0.250000	0.150000
O83	9.560	4.541	2.083153	0.250000	0.150000
O84	9.839	4.537	2.083158	0.250000	0.150000

ID	CG s	Re-setup s	Compliance	Volume	Change
O85	18.021	4.753	2.083156	0.250000	0.150000
O86	10.922	4.575	2.083161	0.250000	0.150000
O87	28.524	3.265	2.370205	0.300000	0.148439
O88	13.261	3.180	2.370205	0.300000	0.148440
O89	12.628	3.158	2.370205	0.300000	0.148435
O90	18.986	3.250	2.370205	0.300000	0.148246
O91	13.677	3.167	2.370205	0.300000	0.148449
O92	17.990	3.165	2.370205	0.300000	0.148433
O93	18.972	2.250	2.375136	0.300000	0.150000
O94	8.998	2.167	2.375136	0.300000	0.150000
O95	8.635	2.146	2.375136	0.300000	0.150000
O96	12.988	2.247	2.375136	0.300000	0.150000
O97	9.270	2.160	2.375137	0.300000	0.150000
O98	9.435	2.174	2.375137	0.300000	0.150000
O99	7.447	6.174	1.942433	0.299999	0.150000
O100	9.530	6.208	1.942433	0.299999	0.150000
O101	22.182	6.392	1.942433	0.299999	0.150000
O102	7.066	6.176	1.942433	0.299999	0.150000
O103	7.795	6.196	1.942433	0.299999	0.150000
O104	12.044	6.394	1.942433	0.299999	0.150000
O105	88.194	8.628	5.774663	0.500001	0.150000
O106	36.427	8.461	5.774663	0.500001	0.150000
O107	28.297	8.447	5.774663	0.500001	0.150000
O108	67.335	8.459	5.774660	0.499999	0.150000
O109	53.178	8.436	5.774661	0.499999	0.150000
O110	88.926	8.634	5.774663	0.500000	0.150000
O111	12.403	4.518	2.083153	0.250000	0.150000
O112	30.396	4.711	2.083154	0.250000	0.150000
O113	9.456	4.497	2.083157	0.250000	0.150000
O114	9.757	4.493	2.083158	0.250000	0.150000
O115	17.767	4.710	2.083157	0.250000	0.150000
O116	10.832	4.524	2.083154	0.250000	0.150000
O117	28.355	3.229	2.370205	0.300000	0.148440
O118	13.447	3.147	2.370205	0.300000	0.148439
O119	12.929	3.133	2.370205	0.300000	0.148439
O120	20.604	3.227	2.370205	0.300000	0.148403
O121	14.123	3.136	2.370205	0.300000	0.148416
O122	13.867	3.124	2.370205	0.300000	0.148421

The following paired ratios divide fused-ai wall time by parity wall time within a repetition. L rows have three ratios; C rows have one. Work=yes requires total Krylov counts within 3% in every pair. Work=no rows remain descriptive and are not qualified fixed-work speedup claims. Median-threshold decisions additionally require the stated compliance/volume/solve gates; these ranges are not bootstrap intervals.

Table 3: All paired wall-time ratios, their observed ranges and Krylov work qualification.

Host	Case	Sch.	Kind	n	Median	Min	Max	Work
G1	C216	64	L	3	1.0096	1.0051	1.0109	yes
G1	C216	32	L	3	0.9989	0.8134	1.0113	no
G1	C512	64	L	3	1.0169	1.0137	1.0265	yes
G1	C512	32	L	3	1.0018	0.9928	1.0129	no
G1	M514	64	L	3	1.0677	1.0668	1.0760	yes
G1	M514	32	L	3	1.2545	0.9962	1.2743	no
G1	T499	64	L	3	1.0491	1.0435	1.0493	no
G1	T499	32	L	3	1.0157	1.0121	1.0200	yes
G1	C216	64	C	1	1.0072	1.0072	1.0072	yes
G1	C216	32	C	1	1.0127	1.0127	1.0127	yes
G2	C216	64	L	3	0.9989	0.9929	1.0145	yes
G2	C216	32	L	3	0.9877	0.9803	1.2157	no
G2	C512	64	L	3	1.0116	1.0040	1.0490	yes
G2	C512	32	L	3	1.0253	0.9761	1.0338	yes
G2	M514	64	L	3	1.0711	1.0388	1.0756	yes
G2	M514	32	L	3	1.4902	1.2104	3.9698	no
G2	T499	64	L	3	1.0336	1.0217	2.1808	yes
G2	T499	32	L	3	1.0051	0.2940	1.0141	no
G2	C216	64	C	1	1.0111	1.0111	1.0111	yes
G2	C216	32	C	1	0.9547	0.9547	0.9547	yes
H1	C216	64	L	3	1.0256	1.0228	1.0289	yes
H1	C216	32	L	3	1.0157	1.0117	1.1832	no
H1	C512	64	L	3	1.0883	1.0870	1.0900	yes
H1	C512	32	L	3	1.0231	1.0207	1.0326	yes
H1	M514	64	L	3	1.1815	1.1694	1.1833	yes
H1	M514	32	L	3	1.1050	0.9158	1.3073	no
H1	T499	64	L	3	1.1187	1.1141	1.1330	yes
H1	T499	32	L	3	1.0402	1.0335	1.0464	yes
H1	C216	64	C	1	1.0209	1.0209	1.0209	yes
H1	C216	32	C	1	0.8505	0.8505	0.8505	no
H2	C216	64	L	3	1.0231	1.0211	1.0292	yes
H2	C216	32	L	3	1.0151	1.0109	1.0489	no
H2	C512	64	L	3	1.0809	1.0800	1.0857	yes
H2	C512	32	L	3	1.0252	1.0237	1.0285	yes
H2	M514	64	L	3	1.1666	1.1645	1.1668	yes
H2	M514	32	L	3	1.1583	1.0959	1.2019	no
H2	T499	64	L	3	1.1151	1.0999	1.1348	yes
H2	T499	32	L	3	1.0469	1.0277	1.0570	no
H2	C216	64	C	1	1.0216	1.0216	1.0216	yes
H2	C216	32	C	1	1.0101	1.0101	1.0101	yes

Scale: every rediscretized solver configuration

n=2 warm tolerance solves per row; all passed the independent physical check. Elements and total/free DOFs follow the recorded grid and x=0 clamp. Both repetition times remain in CSV. The table median is not a total setup-plus-solve time. All six arms were resident in the measured harness. Live-array and transient peak memory are NR for every row. R IDs match the timing split below.

Table 4: Rediscretized scale dimensions and warm time to the requested residual tolerance.

ID	Host	Case	Arm	Elements	DOFs	Free DOFs	Lvls	K	Warm s
R1	IA	q96x1	32P	1769472	5447811	5419584	5	24	0.3283
R2	IA	q96x1	64N	1769472	5447811	5419584	5	24	0.3626
R3	IA	q96x1	64F	1769472	5447811	5419584	5	24	0.3947
R4	IA	q96x1	32N	1769472	5447811	5419584	5	24	0.3196
R5	IA	q96x1	64P	1769472	5447811	5419584	5	24	0.4345
R6	IA	q96x1	32F	1769472	5447811	5419584	5	24	0.3161
R7	IA	q64x2	32F	4194304	12830211	12780288	5	14	0.3153
R8	IA	q64x2	64F	4194304	12830211	12780288	5	14	0.5253
R9	IA	q64x2	64N	4194304	12830211	12780288	5	14	0.4910
R10	IA	q64x2	32P	4194304	12830211	12780288	5	14	0.3323
R11	IA	q64x2	32N	4194304	12830211	12780288	5	14	0.3101
R12	IA	q64x2	64P	4194304	12830211	12780288	5	14	0.5816
R13	IA	q96x2	64N	14155776	43022595	42910848	6	24	2.6556
R14	IA	q96x2	64P	14155776	43022595	42910848	6	24	3.1684
R15	IA	q96x2	64F	14155776	43022595	42910848	6	24	2.8056
R16	IA	q96x2	32P	14155776	43022595	42910848	6	24	1.7375
R17	IA	q96x2	32F	14155776	43022595	42910848	6	24	1.6675
R18	IA	q96x2	32N	14155776	43022595	42910848	6	24	1.6241
R19	IA	q96x3	64F	47775744	144574851	144324288	6	20	7.8558
R20	IA	q96x3	64N	47775744	144574851	144324288	6	20	7.4140
R21	IA	q96x3	32N	47775744	144574851	144324288	6	20	4.5369
R22	IA	q96x3	32P	47775744	144574851	144324288	6	20	4.8427
R23	IA	q96x3	32F	47775744	144574851	144324288	6	20	4.6118
R24	IA	q96x3	64P	47775744	144574851	144324288	6	20	8.9015
R25	IH	q96x1	32P	1769472	5447811	5419584	5	24	0.1952
R26	IH	q96x1	64N	1769472	5447811	5419584	5	24	0.2579
R27	IH	q96x1	64F	1769472	5447811	5419584	5	24	0.2756
R28	IH	q96x1	32N	1769472	5447811	5419584	5	24	0.1872
R29	IH	q96x1	64P	1769472	5447811	5419584	5	24	0.3001
R30	IH	q96x1	32F	1769472	5447811	5419584	5	24	0.1903
R31	IH	q64x2	32F	4194304	12830211	12780288	5	14	0.2457
R32	IH	q64x2	64F	4194304	12830211	12780288	5	14	0.3614
R33	IH	q64x2	64N	4194304	12830211	12780288	5	14	0.3401
R34	IH	q64x2	32P	4194304	12830211	12780288	5	14	0.2452
R35	IH	q64x2	32N	4194304	12830211	12780288	5	14	0.2360
R36	IH	q64x2	64P	4194304	12830211	12780288	5	14	0.3932

ID	Host	Case	Arm	Elements	DOFs	Free DOFs	Lvls	K	Warm s
R37	IH	q96x2	64N	14155776	43022595	42910848	6	24	2.1996
R38	IH	q96x2	64P	14155776	43022595	42910848	6	24	2.4101
R39	IH	q96x2	64F	14155776	43022595	42910848	6	24	2.2341
R40	IH	q96x2	32P	14155776	43022595	42910848	6	24	1.4375
R41	IH	q96x2	32F	14155776	43022595	42910848	6	24	1.4042
R42	IH	q96x2	32N	14155776	43022595	42910848	6	24	1.3919
R43	IH	q96x3	64F	47775744	144574851	144324288	6	20	6.3951
R44	IH	q96x3	64N	47775744	144574851	144324288	6	20	6.1909
R45	IH	q96x3	32N	47775744	144574851	144324288	6	20	3.9780
R46	IH	q96x3	32P	47775744	144574851	144324288	6	20	4.0944
R47	IH	q96x3	32F	47775744	144574851	144324288	6	20	4.0011
R48	IH	q96x3	64P	47775744	144574851	144324288	6	20	6.8738
R49	IR	q96x1	32P	1769472	5447811	5419584	5	24	0.3096
R50	IR	q96x1	64N	1769472	5447811	5419584	5	24	0.8753
R51	IR	q96x1	64F	1769472	5447811	5419584	5	24	0.7545
R52	IR	q96x1	32N	1769472	5447811	5419584	5	24	0.3764
R53	IR	q96x1	64P	1769472	5447811	5419584	5	24	0.5447
R54	IR	q96x1	32F	1769472	5447811	5419584	5	24	0.3524
R55	IR	q64x2	32F	4194304	12830211	12780288	5	14	0.4283
R56	IR	q64x2	64F	4194304	12830211	12780288	5	14	1.0823
R57	IR	q64x2	64N	4194304	12830211	12780288	5	14	1.2460
R58	IR	q64x2	32P	4194304	12830211	12780288	5	14	0.3713
R59	IR	q64x2	32N	4194304	12830211	12780288	5	14	0.4623
R60	IR	q64x2	64P	4194304	12830211	12780288	5	14	0.7927
R61	IR	q96x2	64N	14155776	43022595	42910848	6	24	7.1734
R62	IR	q96x2	64P	14155776	43022595	42910848	6	24	4.5936
R63	IR	q96x2	64F	14155776	43022595	42910848	6	24	6.2614
R64	IR	q96x2	32P	14155776	43022595	42910848	6	24	2.1160
R65	IR	q96x2	32F	14155776	43022595	42910848	6	24	2.4510
R66	IR	q96x2	32N	14155776	43022595	42910848	6	24	2.6349

Shared setup is once per case: reference hierarchy, host coarsest assembly/Cholesky, transfers and spectral estimation. Arm setup is the incremental recorded construction with cached compilation possible. Shared values are repeated for lookup, not additive across arms. Noncoarsest levels are matrix-free; the coarsest operator is assembled and factored.

Table 5: Scale setup and observed independent residual (R IDs as above).

ID	Shared setup s	Arm setup s	Max. residual	Live bytes	Peak bytes
R1	4.6366	0.0030	4.37e-09	NR	NR
R2	4.6366	0.0038	4.37e-09	NR	NR
R3	4.6366	0.0023	4.37e-09	NR	NR
R4	4.6366	0.5481	4.37e-09	NR	NR
R5	4.6366	0.0032	4.37e-09	NR	NR

ID	Shared setup s	Arm setup s	Max. residual	Live bytes	Peak bytes
R6	4.6366	0.0032	4.37e-09	NR	NR
R7	2.1373	0.0036	2.06e-09	NR	NR
R8	2.1373	0.0031	2.06e-09	NR	NR
R9	2.1373	0.0034	2.06e-09	NR	NR
R10	2.1373	0.0034	2.06e-09	NR	NR
R11	2.1373	0.0034	2.06e-09	NR	NR
R12	2.1373	0.0034	2.06e-09	NR	NR
R13	3.4164	0.0053	4.43e-09	NR	NR
R14	3.4164	0.0057	4.43e-09	NR	NR
R15	3.4164	0.0053	4.43e-09	NR	NR
R16	3.4164	0.0041	4.43e-09	NR	NR
R17	3.4164	0.0041	4.43e-09	NR	NR
R18	3.4164	0.0041	4.43e-09	NR	NR
R19	13.6777	0.0173	7.45e-09	NR	NR
R20	13.6777	0.0173	7.45e-09	NR	NR
R21	13.6777	0.0132	7.45e-09	NR	NR
R22	13.6777	0.0132	7.45e-09	NR	NR
R23	13.6777	0.0132	7.45e-09	NR	NR
R24	13.6777	0.0173	7.45e-09	NR	NR
R25	3.3143	0.0016	4.37e-09	NR	NR
R26	3.3143	0.0020	4.37e-09	NR	NR
R27	3.3143	0.0013	4.37e-09	NR	NR
R28	3.3143	0.3885	4.37e-09	NR	NR
R29	3.3143	0.0017	4.37e-09	NR	NR
R30	3.3143	0.0017	4.37e-09	NR	NR
R31	1.5209	0.0018	2.06e-09	NR	NR
R32	1.5209	0.0016	2.06e-09	NR	NR
R33	1.5209	0.0019	2.06e-09	NR	NR
R34	1.5209	0.0019	2.06e-09	NR	NR
R35	1.5209	0.0017	2.06e-09	NR	NR
R36	1.5209	0.0018	2.06e-09	NR	NR
R37	2.2360	0.0035	4.43e-09	NR	NR
R38	2.2360	0.0035	4.43e-09	NR	NR
R39	2.2360	0.0035	4.43e-09	NR	NR
R40	2.2360	0.0031	4.43e-09	NR	NR
R41	2.2360	0.0031	4.43e-09	NR	NR
R42	2.2360	0.0031	4.43e-09	NR	NR
R43	9.5709	0.0112	7.45e-09	NR	NR
R44	9.5709	0.0112	7.45e-09	NR	NR
R45	9.5709	0.0109	7.45e-09	NR	NR
R46	9.5709	0.0108	7.45e-09	NR	NR
R47	9.5709	0.0108	7.45e-09	NR	NR
R48	9.5709	0.0114	7.45e-09	NR	NR

ID	Shared setup s	Arm setup s	Max. residual	Live bytes	Peak bytes
R49	5.1525	0.0024	4.37e-09	NR	NR
R50	5.1525	0.0035	4.37e-09	NR	NR
R51	5.1525	0.0023	4.37e-09	NR	NR
R52	5.1525	0.5727	4.37e-09	NR	NR
R53	5.1525	0.0026	4.37e-09	NR	NR
R54	5.1525	0.0027	4.37e-09	NR	NR
R55	3.3784	0.0029	2.06e-09	NR	NR
R56	3.3784	0.0026	2.06e-09	NR	NR
R57	3.3784	0.0029	2.06e-09	NR	NR
R58	3.3784	0.0029	2.06e-09	NR	NR
R59	3.3784	0.0027	2.06e-09	NR	NR
R60	3.3784	0.0029	2.06e-09	NR	NR
R61	7.4763	0.0080	4.43e-09	NR	NR
R62	7.4763	0.0081	4.43e-09	NR	NR
R63	7.4763	0.0080	4.43e-09	NR	NR
R64	7.4763	0.0051	4.43e-09	NR	NR
R65	7.4763	0.0052	4.43e-09	NR	NR
R66	7.4763	0.0052	4.43e-09	NR	NR

Galerkin capacity and capped solves

Session F uses four-level Galerkin with the arms/schedules shown. An OK row denotes allocation/execution success, not convergence: each retained 40-iteration solve was capped. Device used is CUDA total-minus-free; pool reserved includes allocator caches. Live arrays and construction peaks are NR. The 2 M success and 5 M failure are observations; 3 M is only an estimate. The 2 M grid is 2,000,376 elements / 6,169,152 total DOFs / 6,144,768 free DOFs. The 5 M grid is 5,029,452 / 15,397,956 / 15,353,064.

Table 6: Cold feasibility setup and capped solve; all successful rows have K=40 and did not converge.

Host	Arm	Sch.	Size	Status	K	Setup s	Solve s	Used GiB	Pool GiB
FA	S	64	2M	OK	40	24.240	2.391	46.326	46.267
FA	S	64	5M	OOM	NR	NR	NR	NR	NR
FA	P	64	2M	OK	40	23.083	1.359	45.555	45.507
FA	P	64	5M	OOM	NR	NR	NR	NR	NR
FA	S	32	2M	OK	40	23.852	1.659	46.373	46.313
FA	S	32	5M	OOM	NR	NR	NR	NR	NR
FH	S	64	2M	OK	40	37.495	1.532	46.383	46.266
FH	S	64	5M	OOM	NR	NR	NR	NR	NR
FH	P	64	2M	OK	40	36.674	0.869	45.557	45.506
FH	P	64	5M	OOM	NR	NR	NR	NR	NR
FH	S	32	2M	OK	40	36.519	1.058	46.430	46.312
FH	S	32	5M	OOM	NR	NR	NR	NR	NR

Host	Arm	Sch.	Size	Status	K	Setup s	Solve s	Used GiB	Pool GiB
FR	S	64	2M	OOM	NR	NR	NR	NR	NR
FR	P	64	2M	OOM	NR	NR	NR	NR	NR
FR	S	32	2M	OOM	NR	NR	NR	NR	NR

W2a below records every block and arm. Each block has three fixed 50-iteration solves; the table shows their median. The separate tolerance attempt occurs only in block 0; NR in block 1 is not a failure. The 1 M grid has 1,000,000 elements / 3,106,053 total DOFs / 3,090,600 free DOFs; the 2 M grid has 2,000,376 / 6,169,152 / 6,144,768. Both use four levels. Setup brackets density-dependent setup only; construction before it is excluded.

Table 7: Galerkin W2a fixed-work and tolerance-attempt outcomes; used memory is a post-setup sample, not peak.

Host	Size	Sch.	Arm	Block	Setup s	50 iter. s	Tol. K	Conv.	Tol. s	Used GiB
IA	1M	64	F	0	6.483	2.161	1000	no	44.298	23.499
IA	1M	64	P	0	0.090	2.263	1000	no	44.108	23.524
IA	1M	64	N	0	0.087	2.137	1000	no	43.328	23.501
IA	1M	64	S	0	0.264	2.767	1000	no	54.062	23.915
IA	1M	64	P	1	0.099	2.189	NR	NR	NR	23.532
IA	1M	64	F	1	0.104	2.173	NR	NR	NR	23.509
IA	1M	64	S	1	0.131	2.686	NR	NR	NR	23.915
IA	1M	64	N	1	0.094	2.152	NR	NR	NR	23.509
IA	1M	32	N	0	0.093	2.145	1000	no	44.243	23.532
IA	1M	32	P	0	0.096	2.152	1000	no	43.970	23.556
IA	1M	32	F	0	0.091	2.196	1000	no	43.456	23.532
IA	1M	32	S	0	0.126	2.336	1000	no	47.000	23.938
IA	1M	32	P	1	0.093	2.156	NR	NR	NR	23.556
IA	1M	32	N	1	0.089	2.200	NR	NR	NR	23.532
IA	1M	32	S	1	0.127	2.477	NR	NR	NR	23.938
IA	1M	32	F	1	0.091	2.144	NR	NR	NR	23.532
IA	2M	64	P	0	0.151	2.287	1000	no	46.096	45.974
IA	2M	64	F	0	0.152	2.284	1000	no	46.732	45.927
IA	2M	64	S	0	0.212	3.363	1000	no	67.533	46.737
IA	2M	64	N	0	0.144	2.234	1000	no	45.302	45.927
IA	2M	64	P	1	0.151	2.346	NR	NR	NR	45.974
IA	2M	64	S	1	0.209	3.323	NR	NR	NR	46.737
IA	2M	64	F	1	0.152	2.311	NR	NR	NR	45.927
IA	2M	64	N	1	0.149	2.279	NR	NR	NR	45.927
IA	2M	32	P	0	0.158	2.329	1000	no	45.701	46.020
IA	2M	32	F	0	0.148	2.231	1000	no	45.129	45.974
IA	2M	32	N	0	0.150	2.265	1000	no	45.514	45.974
IA	2M	32	S	0	0.212	2.669	1000	no	54.222	46.784
IA	2M	32	F	1	0.149	2.289	NR	NR	NR	45.974
IA	2M	32	N	1	0.145	2.233	NR	NR	NR	45.974
IA	2M	32	S	1	0.212	2.727	NR	NR	NR	46.784

Host	Size	Sch.	Arm	Block	Setup s	50 iter. s	Tol. K	Conv.	Tol. s	Used GiB
IA	2M	32	P	1	0.151	2.308	NR	NR	NR	46.020
IH	1M	64	F	0	4.450	1.260	1000	no	25.382	23.600
IH	1M	64	P	0	0.072	1.254	1000	no	25.181	23.624
IH	1M	64	N	0	0.071	1.243	1000	no	24.893	23.600
IH	1M	64	S	0	0.181	1.618	1000	no	32.356	24.069
IH	1M	64	P	1	0.072	1.260	NR	NR	NR	23.686
IH	1M	64	F	1	0.072	1.254	NR	NR	NR	23.663
IH	1M	64	S	1	0.095	1.616	NR	NR	NR	24.069
IH	1M	64	N	1	0.071	1.248	NR	NR	NR	23.663
IH	1M	32	N	0	0.069	1.265	1000	no	25.465	23.686
IH	1M	32	P	0	0.071	1.253	1000	no	25.559	23.710
IH	1M	32	F	0	0.070	1.255	1000	no	25.165	23.686
IH	1M	32	S	0	0.096	1.387	1000	no	27.772	24.093
IH	1M	32	P	1	0.072	1.255	NR	NR	NR	23.710
IH	1M	32	N	1	0.071	1.248	NR	NR	NR	23.686
IH	1M	32	S	1	0.097	1.378	NR	NR	NR	24.093
IH	1M	32	F	1	0.071	1.250	NR	NR	NR	23.686
IH	2M	64	P	0	0.129	1.361	1000	no	27.593	46.126
IH	2M	64	F	0	0.126	1.362	1000	no	27.139	46.079
IH	2M	64	S	0	0.175	2.156	1000	no	42.246	46.889
IH	2M	64	N	0	0.124	1.346	1000	no	27.031	46.079
IH	2M	64	P	1	0.127	1.361	NR	NR	NR	46.126
IH	2M	64	S	1	0.175	2.105	NR	NR	NR	46.889
IH	2M	64	F	1	0.130	1.359	NR	NR	NR	46.079
IH	2M	64	N	1	0.127	1.347	NR	NR	NR	46.079
IH	2M	32	P	0	0.130	1.345	1000	no	26.789	46.173
IH	2M	32	F	0	0.126	1.373	1000	no	27.031	46.126
IH	2M	32	N	0	0.125	1.331	1000	no	26.659	46.126
IH	2M	32	S	0	0.175	1.620	1000	no	32.364	46.936
IH	2M	32	F	1	0.128	1.339	NR	NR	NR	46.126
IH	2M	32	N	1	0.123	1.334	NR	NR	NR	46.126
IH	2M	32	S	1	0.176	1.623	NR	NR	NR	46.936
IH	2M	32	P	1	0.129	1.339	NR	NR	NR	46.173

Hardware/software and experiment manifest

The synthetic rows S4090/S3090 precede Sessions A–I and directly measured only plain parity. H100 NVL (FH) and H100 80GB HBM3 (IH) are distinct. Every A100 measured here is PCIe; the historical A100 SXM roofline specification is not measured hardware. CPU NR means the record did not retain a model name. Every row is one allocation.

Table 8: Measured allocation identities; complete filenames and machine identifiers are in experiment-manifest.csv.

Key	GPU	Capacity	CPU
A	NVIDIA GeForce RTX 4090	24564 MiB	NR
B	NVIDIA A100 80GB PCIe	81920 MiB	Xeon® Gold 5218R
C	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 7B13 64-Core Processor
D	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 9684X 96-Core Processor
E1	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 7742 64-Core Processor
E2	NVIDIA GeForce RTX 4090	24564 MiB	AMD Ryzen Threadripper 2950X 16-Core Processor
FA	NVIDIA A100 80GB PCIe	81920 MiB	AMD Ryzen Threadripper PRO 5955WX 16-Cores
FH	NVIDIA H100 NVL	95830 MiB	XEON® PLATINUM 8558
FR	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 7R13 48-Core Processor
G1	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 7282 16-Core Processor
G2	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 7702 64-Core Processor
H1	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 7502 32-Core Processor
H2	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 7502 32-Core Processor
IA	NVIDIA A100 80GB PCIe	81920 MiB	AMD EPYC 7713 64-Core Processor
IH	NVIDIA H100 80GB HBM3	81559 MiB	AMD EPYC 9454 48-Core Processor
IR	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 7702 64-Core Processor
S4090	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 7B13 64-Core Processor
S3090	NVIDIA GeForce RTX 3090	24576 MiB	AMD EPYC 7532 32-Core Processor
J	NVIDIA GeForce RTX 4090	24564 MiB	AMD EPYC 7K62 48-Core Processor
CEED	NVIDIA GeForce RTX 4090	24564 MiB	AMD Ryzen 5 5600X 6-Core Processor

Table 9: Software from installation records. Installed and loaded-version queries are distinguished. Native libCEED nvcc/GCC versions come from retained version-command output.

Key	Driver	CUDA runtime	CuPy	CUDA/host compiler
A	580.173.02	12090 (linked to CuPy) / 12080 (locally installed)	13.6.0	NVRTC (12, 8)
B	595.84	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
C	590.48.01	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
D	580.119.02	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
E1	580.95.05	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
E2	595.71.05	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
FA	580.126.09	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)

Key	Driver	CUDA runtime	CuPy	CUDA/host compiler
FH	570.211.01	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
FR	595.84	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
G1	570.133.07	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
G2	580.95.05	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
H1	560.35.03	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
H2	570.211.01	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
IA	595.71.05	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
IH	595.71.05	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
IR	560.35.03	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
S4090	595.84	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
S3090	595.84	12090 (linked to CuPy) / 12060 (locally installed)	13.6.0	NVRTC (12, 6)
J	535.183.01	12.2.140 installed cudart; linked version NR	13.6.0	NVRTC 12.2.140
CEED	595.58.03	CUDA 12.6 SDK/header probe	native harness	nvcc 12.6.85; GCC 11.4.0

Precision, kernels, sizes, repeated work and timing-boundary definitions are explicit in the Timing, optimization outcomes and measured scale narrative and in the structured manifest CSV. In particular: synthetic products use five warmups and 11 rounds of 20 calls; A–C use 11 product rounds, 10 fixed-work rounds, and 3/3/2 tolerance repetitions; D/E use 5 tolerance and 10 fixed-work repetitions (E has 11 product rounds); G/H use 3 paired trajectories plus one capped trajectory per arm/schedule; I uses 5 product rounds of 10 calls, 2 tolerance repetitions and 3 fixed-work rounds. G/H warm one full step per arm; A–C warm one FP64 PCG iteration or FP32 action; D/E/I warm two solver iterations per arm. Setup, transfers, masks and zeroing depend on the endpoint as specified in the narrative, not on a universal timer boundary.